CHAPTER 9

THE SOIL-WATER BALANCE AND CLIMATE

A basic concept of physical geography is that the availability of water to plants is a more important factor in the environment than precipitation itself. Much of the water received as precipitation is lost in a variety of ways and is not usable for plants. Just as in a fiscal budget, when the monthly or yearly loss of moisture exceeds the precipitation, a budgetary deficit results; when precipitation exceeds the losses, a budgetary surplus results. An analysis of the water budget of a given area of the earth's surface is actually arrived at in much the same way as a fiscal budget. The accounting requires only additions and subtractions of amounts within fixed periods of time, such as the calendar month or year.

To understand why such moisture gains and losses occur, we need to study the physical processes affecting water in its vapor, liquid, and solid states—not only in the atmosphere but also in the soil and rock and in the exposed water of streams, lakes, and glaciers. The science of *hydrology* treats such relationships of water as a complex but unified system on the earth. The primary concern of this chapter is the hydrology of the soil zone.

SURFACE AND SUBSURFACE WATER

We classify water on the land according to whether it is *surface water,* flowing exposed or ponded on the land, or *subsurface water,* occupying openings in the soil or rock. Water held in the soil within a meter or two of the surface is the *soil water;* it is the particular concern of the plant ecologist, soil scientist, and agricultural engineer. Water held in the openings of the bedrock is referred to as ground water and is studied by the geologist. Surface water and ground water are the subjects of Chapter 16.

THE HYROLOGIC CYCLE AND THE GLOBAL WATER BALANCE

Water of oceans, atmosphere, and lands move in a great series of continuous interchanges of both geographic position and physical state, known as the *hydrologic cycle* (Figure 9.1). A particular molecule of water might, if we could trace it continuously, travel through any one of a number of possible circuits involving alternately the water vapor state and the liquid or solid state.

The pictorial diagram of the hydrologic cycle given in Figure 9.1 can be quantified for the earth as a whole. Figure 9.2 is a mass-flow diagram relating the principal pathways of the water circuit. We can start with the oceans, which are the basic reservoir of free water. Evaporation from the ocean surfaces totals about 419,000 cu km per year (English equivalents are given in Figure 9.2). At the same time, evaporation from soil, plants, and water surfaces of the continents totals about 69,000 cu km. Thus the total evaporation is 488,000 cu km; it represents the quantity of water that must be returned annually to the liquid or solid state.

Precipitation is unevenly divided between continents and oceans; 106,000 cu km is received by the land surfaces and 382,000 cu km by the ocean surfaces. Notice that the continents receive about 37,000 cu km more water as precipitation than they lose by evaporation.

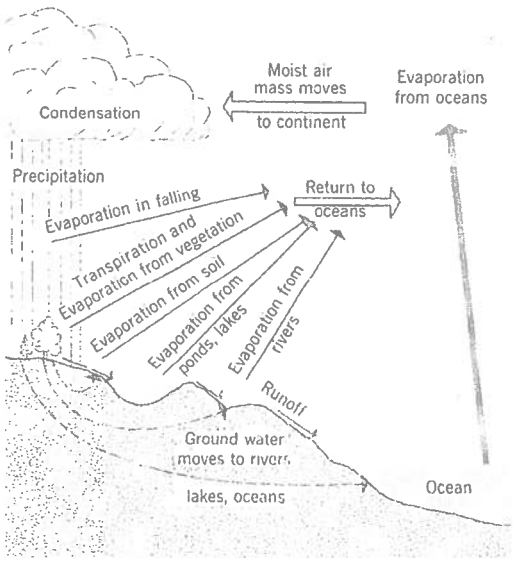


FIGURE 9.1 The hydrologic cycle. (After Holtzman.)

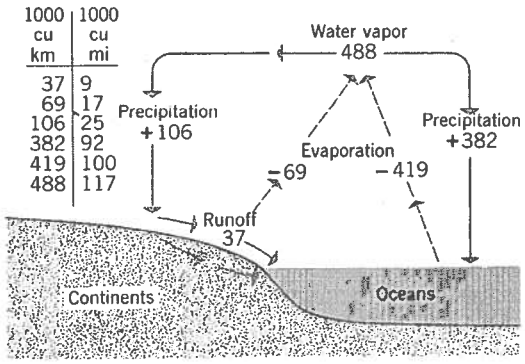


FIGURE 9.2 The global water balance. Figures give average annual water flux in and out of world land areas and world oceans. (Data of John R. Mather, 1974. From A. N. Strahler, *The Earth Sciences*, 2nd ed., Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

This excess quantity flows over or under the ground surface to reach the sea; it is collectively termed *runoff*.

We can state the *global water balance* as follows:

$$P = E + G + R$$

where P = precipitation
E = evaporation
G = net gain or loss of water in the system, a storage term
R = runoff (positive when out of the continents, negative when into the oceans)

All terms have the dimensions of volume per unit time (e.g., cubic kilometers per year). When applied over the span of a year, and averaged over many years, the storage term G can be considered as zero because the global system is essentially closed so far as matter is concerned. The quantities of water in storage in the atmosphere, on the lands, and in the oceans will remain about constant from year to year.

The equation then simplifies to

$$P = E + R$$

Using the figures given for the continents,

$$106{,}000 = 69{,}000 + 37{,}000$$

and for the oceans,

$$382{,}000 = 419{,}000 - 37{,}000$$

For the globe as a whole, combining continents and ocean basins, the runoff terms cancel out:

$$106{,}000 + 382{,}000 = 69{,}000 + 419{,}000$$
$$488{,}000 = 488{,}000$$

THE HYDROLOGIC CYCLE AS A CLOSED MATERIAL FLOW SYSTEM

The global water balance can be treated as a material flow system, using the type of schematic diagram described in Chapter 2. Figure 9.3 shows a closed system representing the total global hydrosphere. Three subsystems are present: atmosphere, continents, and oceans. For the sake of simplicity, glacier ice is included with liquid water as a single form of water. In the atmosphere subsystem the input is in the form of water vapor derived by evaporation from the ocean surface and by evaporation and evapotranspiration from the continental surface. Stored water vapor undergoes condensation to become precipitation in the liquid form as rain (or solid state as snow), leaving the atmosphere subsystem and entering either the continent subsystem or the ocean subsystem. By runoff, water in storage on the continents is transferred to the oceans without change of state.

THE GLOBAL WATER BALANCE BY LATITUDE ZONES

From a global water balance in terms of water volume per unit of time (cu km/yr; cu mi/yr), we now turn to water balance data in terms of water depth per unit of time (cm/yr; in./yr). Whereas the first set of data tells us the total quantity of water transported into or out of a specified area in one year, the second set will tell us the intensity of water flow, independent of total surface area and total water quantity.

The water balance can be estimated for latitude belts of 10° width. This treatment reveals a response to the latitudinal changes in radiation and heat balances from the equatorial zone to the polar zones. Figure 9.4 shows

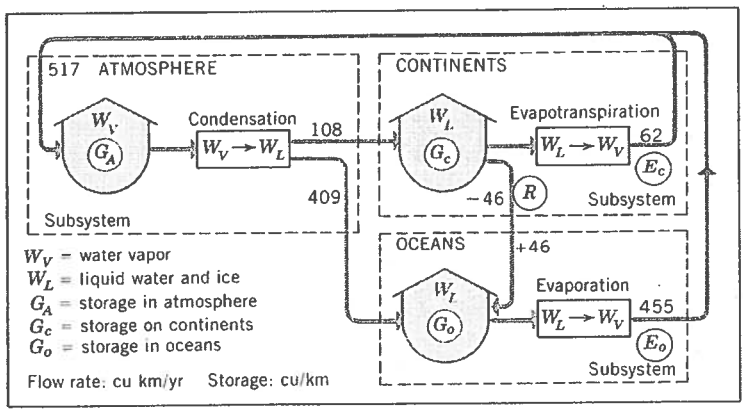


FIGURE 9.3 Schematic diagram of the hydrologic cycle as a closed material system.

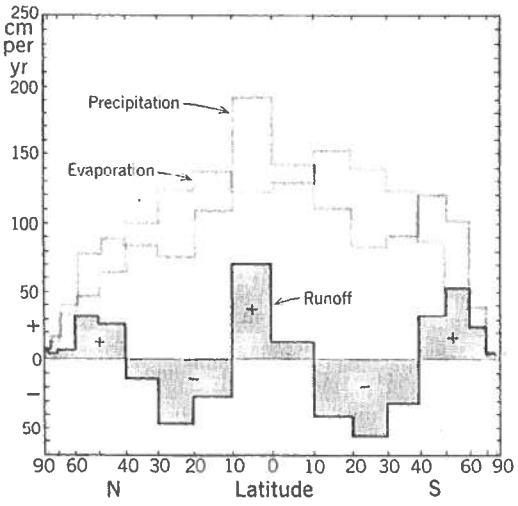


FIGURE 9.4 The water balance for 10° latitude zones. (Data from several sources compiled by W. D. Sellers. From A. N. Strahler, *The Earth Sciences*, 2nd ed., Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

the average annual values of precipitation, evaporation, and runoff. The runoff term R in this case includes import or export of water in or out of the belt by ocean currents as well as by stream flow. Notice the equatorial zone of water surplus with positive values of R, in contrast to the subtropical belts, with an excess of evaporation. The runoff term R, which has a negative sign in the subtropical belts, represents the importation of water by ocean currents to furnish the quantity needed for evaporation. Water surpluses occur poleward of the 40th parallel; but the values of all three terms decline rapidly at high latitudes, going nearly to zero at the poles. This graph should be compared with Figure 6.28, which shows water-vapor transport across parallels of latitude. Notice that the precipitation surpluses are sustained by importation of water vapor and that evaporation surpluses are sustained by export of water vapor.

GLOBAL WATER IN STORAGE

The total global water resource is stored in the gaseous, liquid, and solid states. Water of the oceans constitutes over 97 percent of the total, as we would expect (Figure 9.5). Next comes water in storage in glaciers, a little over 2 percent. Of the remaining quantity, almost all is ground water, so surface water in lakes and streams is a very small quantity indeed. But it is surface water, together with the very small quantity of soil water, that sustains all life of the lands. Some environmentalists consider that fresh surface water will prove to be the limiting factor in the capacity of our planet to support the rapidly expanding human population. The quantity of water vapor held in the atmosphere is also very small—only about 10 times greater than that held in streams; but this atmospheric moisture is the source of all fresh water of the lands.

GLOBAL WATER BALANCE AND CLIMATE CHANGE

Long-term changes in water-balance quantities are associated with atmospheric environmental changes. For example, atmospheric cooling on a global scale would bring a reduction of atmospheric water-vapor storage. This change would, in turn, reduce precipitation and runoff generally. But at the same time, a greater proportion of that precipitation would be in the form of snow, so the storage in ice accumulations would rise and the storage in ocean waters would fall. These changes describe the changing water balance associated with onset of an ice age, or glaciation—a major environmental change already experienced by our planet at least four times in the past two million years (see Chapter 22).

INFILTRATION AND RUNOFF

Most soil surfaces in their undisturbed, natural states are capable of absorbing the water from light or moderate rains. This absorption process is known as *infiltration*. Most soils have natural passageways between poorly fitting soil particles, as well as larger openings, such as earth cracks resulting from soil drying, borings of worms and animals, cavities left from decay of plant roots, or openings made by growth and melting of frost crystals. A mat of decaying leaves and stems breaks the force of falling drops and helps keep these openings clear. If rain falls too rapidly to be passed downward through these soil openings, the excess amount flows as a surface-water

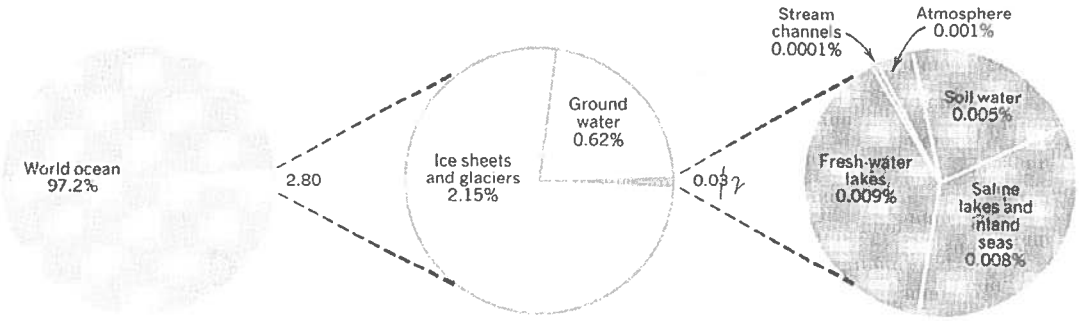


FIGURE 9.5 The total volume of global water in storage is largely held in the world ocean.

layer down the direction of ground slope. This surface runoff, called overland flow, is described in Chapter 16.

As stated in Chapter 6, rainfall is measured in units of centimeters or inches per hour. This is the depth to which water will accumulate in each hour if rain is caught in a flat-bottomed, straight-sided container, assuming none to be lost by evaporation or splashing out (see Figure 16.2). Similarly, infiltration is stated in centimeters or inches per hour and can be thought of as the rate at which the water level in the same container might drop if the water were leaking through a porous base. Runoff, also stated in cm (in.) per hour, may be thought of as the amount of overflow of the container per hour when rain falls too fast to be disposed of by leaking through the base.

EVAPORATION AND TRANSPIRATION

Between periods of rain, water held in the soil is gradually given up by a twofold drying process. First, direct evaporation into the open air occurs at the soil surface and progresses downward. Air also enters the soil freely and may actually be forced alternately in and out of the soil by atmospheric pressure changes. Even if the soil did not "breathe" in this way, there would be a slow diffusion of water vapor surfaceward through the open soil pores. Ordinarily only the upper 30 cm (1 ft) of soil is dried by evaporation in a single dry season. In the prolonged drought of deserts, a dry condition extends to depths of many meters.

Second, plants draw the soil water into their systems through vast networks of tiny rootlets. This water, after being carried upward through the trunk and branches into the leaves, is discharged through leaf pores into the atmosphere in the form of water vapor. The process is termed *transpiration.*

In studies of climatology and hydrology, it is convenient to use the term *evapotranspiration* to cover the combined water loss from direct evaporation and the transpiration of plants. The rate of evapotranspiration slows down as soil-water supply becomes depleted during a dry summer period because plants use various devices to reduce transpiration. In general, the less water remaining, the slower the loss through evapotranspiration.

Figure 9.6 illustrates the various terms explained up to this point and serves to give a more detailed picture of that part of the hydrologic cycle involving the soil. The soil layer from which plants can draw moisture is the *soil-water belt.* This belt gains water through precipitation and infiltration. As the minus signs show, the soil loses water through transpiration, evaporation, and overland flow. Excess water also leaves the soil by downward *gravity percolation* to the ground-water zone below. Between the soil-water belt and the ground-water zone is an *intermediate belt.* Here water is held at a depth too great to be returned to the surface by evapotranspiration because it is below the level of plant roots.

GROUND WATER AND STREAM FLOW

Gravity percolation carries excess water down to the ground-water zone, in which all pore spaces are fully saturated (Figure 9.6). Within this zone water moves slowly in deep paths, eventually emerging by seepage into streams, ponds, and lakes.

Excess water leaves the area by stream flow. Streams are fed directly by overland flow in periods of heavy, prolonged rain or rapid snowmelt. Streams that flow throughout the year—perennial streams—derive much of their water from ground water seepage. Streams fed only by overland flow run intermittently, and their channels are dry much of the time between rain periods. We shall investigate ground-water flow and stream flow in detail in Chapter 16.

WATER IN THE SOIL

When infiltration occurs, the water is drawn downward through the soil pores, wetting successively lower layers. This activity is called *soil-water recharge.* Eventually the soil layer holds the maximum possible quantity of water, although the larger pores remain filled with air. Water movement then continues downward through the underlying intermediate belt.

Suppose now that the rain stops and several days of dry weather follow. Excess soil water continues to drain downward under gravity, but some water clings to the

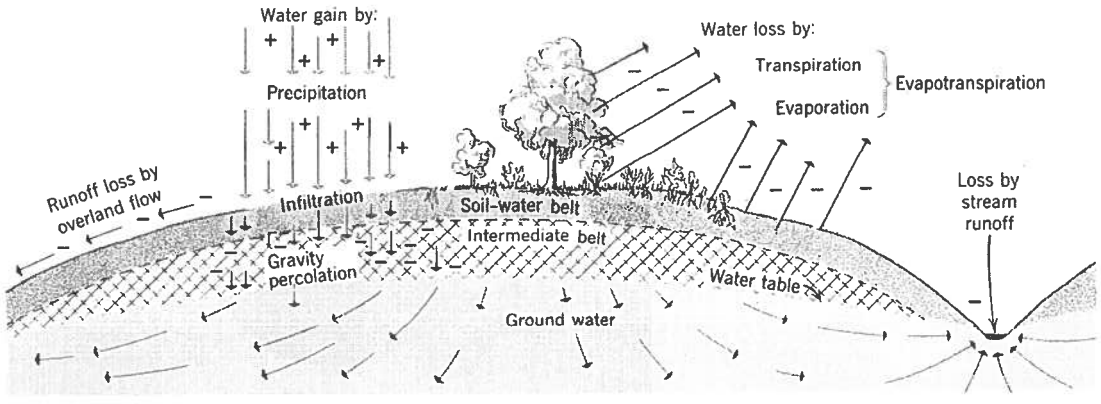


FIGURE 9.6 The soil-water belt occupies an important posi[illegible] hydrologic cycle.

soil particles and effectively resists the pull of gravity because of the force of capillary tension. We are all familiar with the way a water droplet seems to be enclosed in a "skin" of surface molecules, drawing the droplet together into a spherical shape so that it clings to the side of a glass indefinitely without flowing down. Similarly, tiny films of water adhere to the soil grains, particularly at the points of grain contacts, and will stay until disposed of by evaporation or absorption into plant rootlets.

When a soil has first been saturated by water, then allowed to drain under gravity until no more water moves downward, the soil is said to be holding its *storage capacity* of water. (Among soil scientists, storage capacity is referred to as *field capacity*.) Drainage takes no more than two or three days for most soils. Most excess water is drained out within one day. Storage capacity is measured in units of depth, usually centimeters or inches, just as is precipitation. For example, a storage capacity of 2 cm in the uppermost 10 cm of the soil means that for a given cube of soil, 10 cm on a side (1 cu decimeter), all the extractable water would form a layer of water 2 cm deep in a 10 × 10 cm container with flat bottom and vertical sides. This depth would be equivalent to complete absorption of a 2-cm rainfall by a completely dry 10-cm layer of soil.

Storage capacity of a given soil depends largely on its texture. Sandy soil has a very small storage capacity; clay soil has a large capacity.

THE SOIL-WATER CYCLE

We can turn next to consider the annual water budget of the soil. Figure 9.7 shows the annual cycle of soil water for a single year at an agricultural experiment station in Ohio. This example can be considered generally representative of conditions in moist, midlatitude climates where there is a strong temperature contrast between winter and summer. Let us start with the early spring (March). At this time the evaporation rate is low because of low air temperatures. The abundance of melting snows and rains has restored the soil water to a surplus quantity. For two months the quantity of water precolating through the soil and entering the ground water keeps the soil pores nearly filled with water. This is the time of year when one encounters soft, muddy ground conditions, whether driving on dirt roads or walking across country. This, too, is the season when runoff is copious and major floods may be expected on larger streams and rivers. In terms of the soil-water budget, a *water surplus* exists.

By May the rising air temperatures, increasing evaporation, and full growth of plant foliage bring on intense evapotranspiration. Now the soil water falls below the storage capacity, although it may be restored temporarily by unusually heavy rains in some years. By midsummer a large *water deficit* exists in the water budget. Even the occasional heavy thunderstorm rains of summer cannot restore the water lost by heavy evapotranspiration. Small springs and streams dry up, and the soil becomes firm and dry. By November, however, the soil water again begins to increase. This is because the plants go into a dormant state, sharply reducing transpiration losses. At the same time, falling air temperatures reduce evaporation. By late winter, usually in February at this location, the storage capacity of the soil is again fully restored.

THE SOIL-WATER BALANCE

From the example of soil-water change in a single year, we move forward to a more generalized concept. The gain, loss, and storage of soil water are accounted for in the *soil-water balance*. Figure 9.8 is a flow diagram that illustrates the components of the balance. *Soil-water storage* (*S*), the actual quantity of water held in the soil-water zone, is increased by recharge during *precipitation* (*P*) but decreased by use through *evapotranspiration* (*E*). Any

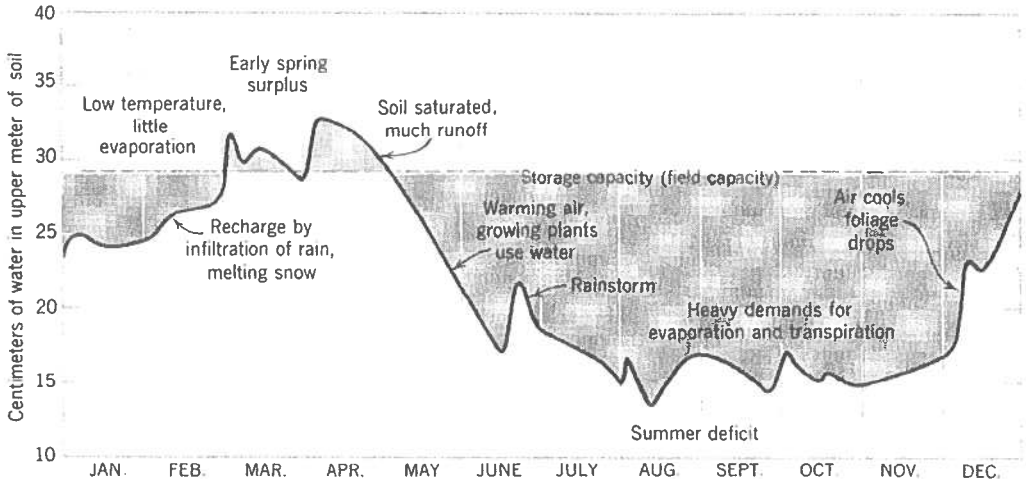


FIGURE 9.7 A typical annual cycle of soil-water change in the Midwest shows a short period of surplus in the spring and a long period of deficit in the summer and fall. (Data of Thornthwaite and Mather.)

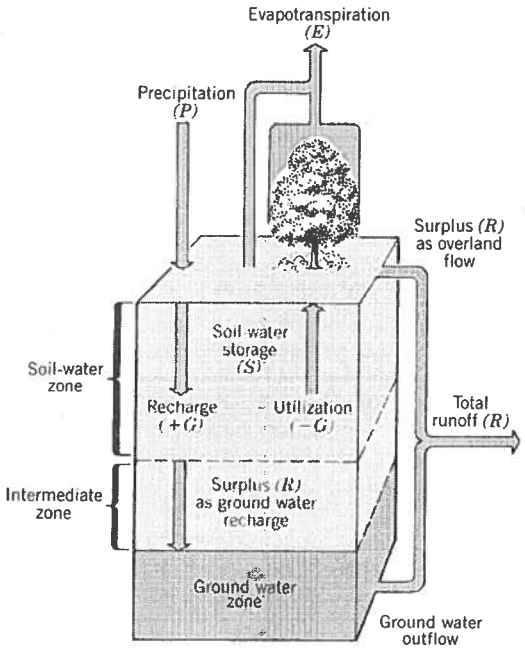


FIGURE 9.8 Schematic diagram of the soil-water balance in a soil column. (From A. N. Strahler, *The Earth Sciences*, 2nd ed., Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

water surplus (*R*) is disposed of by downward percolation to the ground-water zone or by overland flow.

The soil column is assumed to have a unit cross-sectional area, for example, 1 sq cm. We can therefore use flow units of depth per unit time (e.g., cm/month or cm/year).

The water-balance equation is as follows:

$$P = E + G + R$$

where P = precipitation
E = evaporation
G = change in soil-water storage
R = water surplus

To proceed, we must recognize two ways to define evapotranspiration. First is *actual evapotranspiration* (*Ea*), which is the true or real rate of water-vapor return to the atmosphere from the ground and its plant cover. Second is *potential evapotranspiration* (*Ep*), representing the water-vapor flux under an ideal set of conditions. One condition is that there be present a complete (or closed) cover of uniform vegetation consisting of fresh green leaves, and no bare ground exposed through that cover. The leaf cover is assumed to have a uniform height above ground—whether the plants be trees, shrubs, or grasses. A second condition is that there be an adequate water supply, such that the storage capacity of the soil is maintained at all times. This condition can be filled naturally by abundant and frequent precipitation or artificially by irrigation. To simplify the ponderous terms we have just defined, they may be transformed as follows:

"actual evapotranspiration" is *water use* (*Ea*)
"potential evapotranspiration" is *water need* (*Ep*)

The word "need" signifies the quantity of soil water needed if plant growth is to be maximized for the given conditions of solar radiation and air temperature and the available supply of nutrients.

The difference between water use and water need is the *soil-water shortage* (*D*). This is the quantity of water that must be furnished by irrigation to achieve maximum crop growth within an agricultural system.

All the terms in the soil-water budget are stated in centimeters of water depth, the same as for precipitation. (We will not give English equivalents on the following pages.) A particular value of storage capacity of the soil is established in advance. In the soil-water budgets we present here, it has been assumed that the soil layer has a storage capacity of 30 cm (about 12 in.).

Several rigorous methods have been devised for estimating the monthly water need (potential evapotranspiration) at any given location on the globe. The method used in this chapter was developed by C. Warren Thornthwaite, a distinguished climatologist and geographer. Thornthwaite's method is based on air temperature, as well as latitude and date. Latitude and season determine intensity and duration of solar radiation received at the ground. Put another way, water need is a measure of the maximum capability of a land surface to return energy from the surface to the atmosphere by the mechanism of latent heat flow under the defined conditions of plant cover and water supply.

THE SOIL-WATER BALANCE AS AN OPEN MATERIAL FLOW SYSTEM

The soil-water balance can be treated as an open material flow system, as shown in Figure 9.9. System input is by precipitation. Rain or snow falling on plant foliage and stems is intercepted and goes into temporary storage; it may be lost by evaporation (change of state), or it may drip or flow down to the ground surface. Some water is temporarily stored as *surface detention* in depressions on the soil surface, from which it may be lost by evaporation, or may leave the system as overland flow. By infiltration, water enters storage in the soil-water zone. Some of this stored water may evaporate from the soil surface, leaving the system as water vapor. Uptake of water through the roots of plants is another way in which soil-water storage is depleted. Transpiration disposes of this water in the vapor state. Surplus soil water may infiltrate downward through the intermediate zone, entering the ground-water zone and leaving the system as ground-water flow.

A SIMPLE SOIL-WATER BUDGET

A simplified *soil-water budget* for the year is shown in Figure 9.10, a graph on which monthly means are plotted as points and connected by smooth lines. In this example, precipitation (P) is much the same in all months, with no strong yearly cycle. In contrast, water need (Ep) shows a strong seasonal cycle: low values in winter and a high summer peak. For midlatitudes in a moist climate, this model is about right.

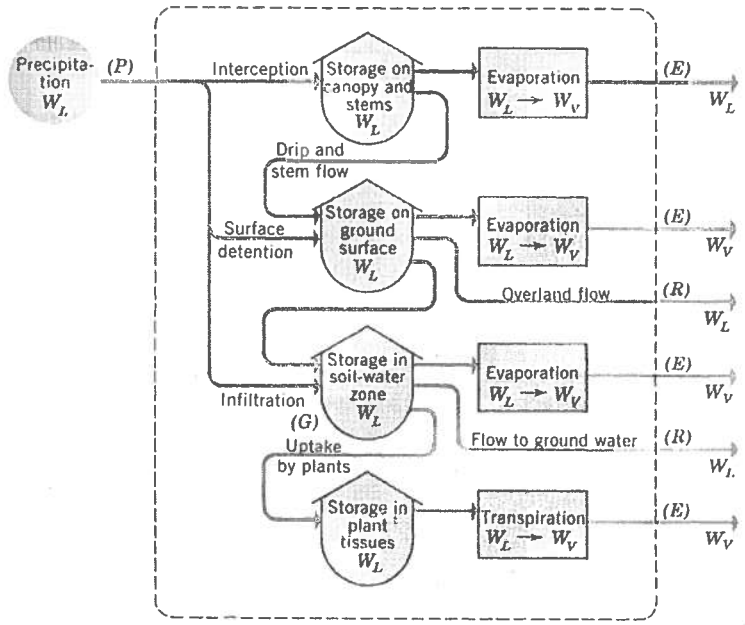


FIGURE 9.9 Schematic diagram of the soil-water balance as an open material flow system.

At the start of the year a large water surplus (R) exists, disposed of by runoff. By May conditions have switched over to a water deficit. In this month plants begin to withdraw soil water from storage. *Storage withdrawal* (−*G*) is represented by the difference between the water-use curve and the precipitation curve (−G is calculated as Ea − P, when Ea exceeds P). As storage withdrawal continues, the plants reduce their water use to less than the optimum quantity, so without irrigation the water-use curve departs from the water-need curve. Storage withdrawal continues throughout the summer. The deficit period lasts through September. The area labeled "soil-water shortage" represents the total quantity of water needed by irrigation to ensure maximum growth throughout the deficit period. Soil-water shortage (D), is calculated by subtracting Ea from Ep (D = Ep − Ea).

In October precipitation (P) again exceeds water need (Ep), but the soil must first absorb an amount equal to the summer storage withdrawal (−G). So we next have a period of *storage recharge* (+*G*); it lasts through November. In December the soil has reached its full storage capacity (30 cm). Now a water surplus (R) again sets in, lasting throughout the winter.

Thornthwaite was concerned with practical problems of crop irrigation. He developed the calculation of the soil-water budget to place crop irrigation on a precise, accurate basis. The Thornthwaite method was later adopted by soil scientists of the U.S. Department of Agriculture to identify classes of soils associated with various soil-water budgets. Water need (potential evapotranspiration) is a difficult quantity to measure, and several other methods have been developed to estimate its true value.

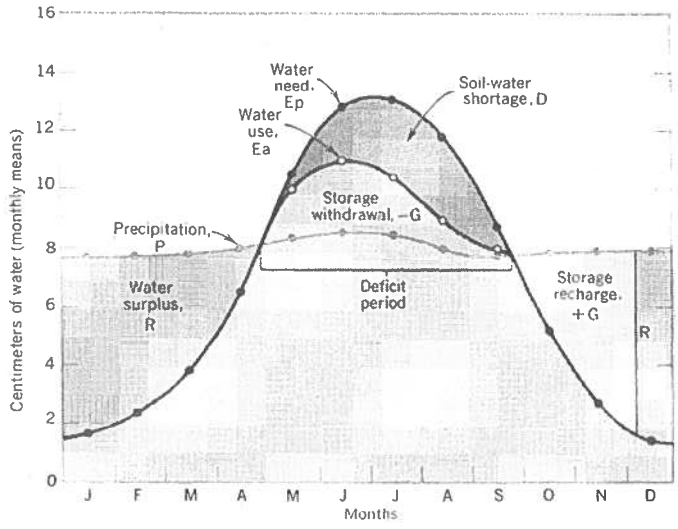


FIGURE 9.10 A simplified soil-water budget typical of a humid midlatitude climate.

TABLE 9.1 **Simplified Example of a Soil-Water Budget**

Equation	P = Ea	+G		+R	Ep	(Ep − Ea) = D
January	11.0 = 1.0			+10.0	1.0	0.0
February	9.0 = 2.0			+ 7.0	2.0	0.0
March	6.0 = 3.5			+ 2.5	3.5	0.0
April	3.0 = 6.0	−3.0			6.0	0.0
May	2.5 = 7.0	−4.5			8.5	1.5
June	2.0 = 6.0	−4.0			9.5	3.5
July	2.5 = 5.0	−2.5			9.0	4.0
August	4.0 = 4.5	−0.5			7.0	2.5
September	7.0 = 4.5		+2.5		4.5	0.0
October	9.0 = 3.0		+6.0		3.0	0.0
November	10.5 = 1.5		+6.0	+ 3.0	1.5	0.0
December	12.0 = 1.5			+10.5	1.5	0.0
Totals	78.5 = 45.5	−14.5	+14.5	+33.0	57.0	11.5
	78.5 = 78.5					

In a world beset by severe and prolonged food shortages, the Thornthwaite concepts and calculations are of great value in assessing the benefits to be gained by increased irrigation. Only in a few parts of the tropical and midlatitude zones is precipitation ample to fulfill the water need during the growing season. In contrast, the equatorial zone generally has a large water surplus throughout the year.

CALCULATING A SIMPLE SOIL-WATER BUDGET

Table 9.1 gives simplified data for a hypothetical example of a soil-water budget. Columns are arranged so that the terms appear in the same sequence as in the soil-water balance equation. Values are rounded to multiples of 0.5 cm to simplify addition and subtraction. Figure 9.11 shows the same data plotted in graphic form.

By means of simple arithmetic, we have calculated the soil-water budget for each month singly and for the year as a whole. Tallied separately at the right are the monthly differences between Ep and Ea, giving a total soil-water shortage, D, of 11.5 cm for the year. This is the quantity of water that would have to be supplied by irrigation in an average year to sustain the full value of Ep. The importance of the water budget calculation in estimating the need for irrigation of crops should be obvious. We have also been able to calculate the annual surplus (R = 33 cm). This information can be used to estimate the recharge of ground water and runoff into streams. In this way an assessment of the water-resource

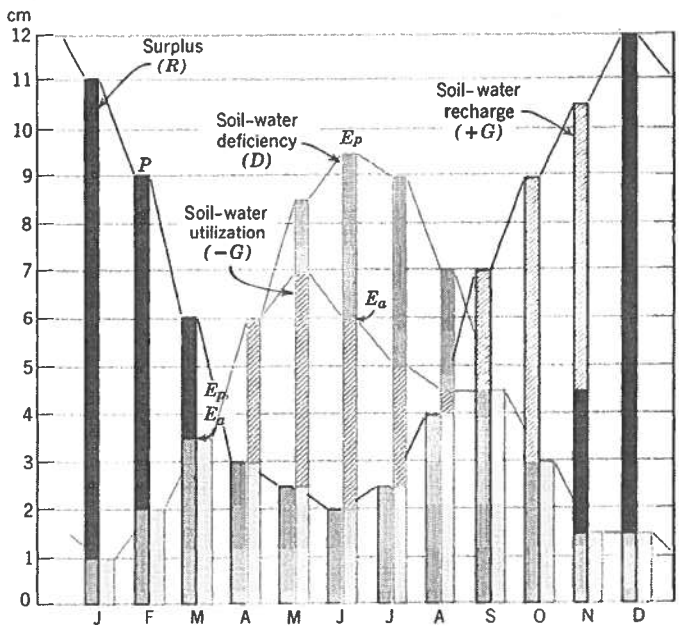


FIGURE 9.11 A model soil-water budget. Bars are correctly scaled to agree with the figures in Table 9.1.

potential of a region can be made—a vital consideration in planning regional economic development and resource management.

THE GLOBAL RANGE OF WATER NEED

Figure 9.12 is a world map showing mean annual water need (Ep, potential evapotranspiration) calculated by the Thornthwaite method. Isopleths of water need in centimeters per year are greatly generalized and are deleted from major highland regions. What counts is the general pattern of poleward decrease of Ep from high values in the equatorial and tropical zones to very low values in the arctic zone. A second important feature is the effect of deserts, which produce abnormally high values compared with areas of moist climate in which temperatures are moderated by precipitation and persistent cloud cover.

Figure 9.13 illustrates the effect of latitude on the annual cycle of water need. Both air-temperature cycles and insolation cycles are reflected in these cycles. Each chart has been labeled as to climate, according to the system given in Chapter 8.

In the wet equatorial climate (1), water need is high all year and the total is over 150 cm (60 in.). For the monsoon and trade-wind littoral climate (2) in the tropical latitude zone, there is a pronounced annual cycle, but a large annual total water need: almost 150 cm (60 in.). In the tropical desert climate (4d), the annual cycle is very strongly developed and the total water need remains high: nearly 127 cm (50 in.). In midlatitudes on the west coast, the Mediterranean climate (7) shows a well-developed annual cycle; but monthly values of water need remain high throughout the mild winter. In the moist continental climate (10) winter months have no water need, but the summer peak is high; the total is 75 cm (30 in.). In the marine west-coast climate (8), close to the Pacific Ocean, however, water need remains substantial throughout the winter because of the mild temperatures. Farther poleward in the boreal forest climate (11), the number of months of zero water need increases to six and the annual total water need diminishes to 40 cm (16 in.). North of the arctic circle, the tundra climate station (12) shows nine consecutive winter months of zero water need and a very narrow summer peak. The total water need here is least of all: 20 cm (8 in.).

The details of these examples are not important. What counts is the trends they show. Total annual water need diminishes from a maximum in the equatorial zone to a minimum in the arctic zone. At the same time, the annual cycle becomes stronger and stronger in the poleward direction. As the cold of winter makes its effects more pronounced, the number of months of zero water need increases. Of course, plant growth is essentially zero when soil water is frozen, so the growing season

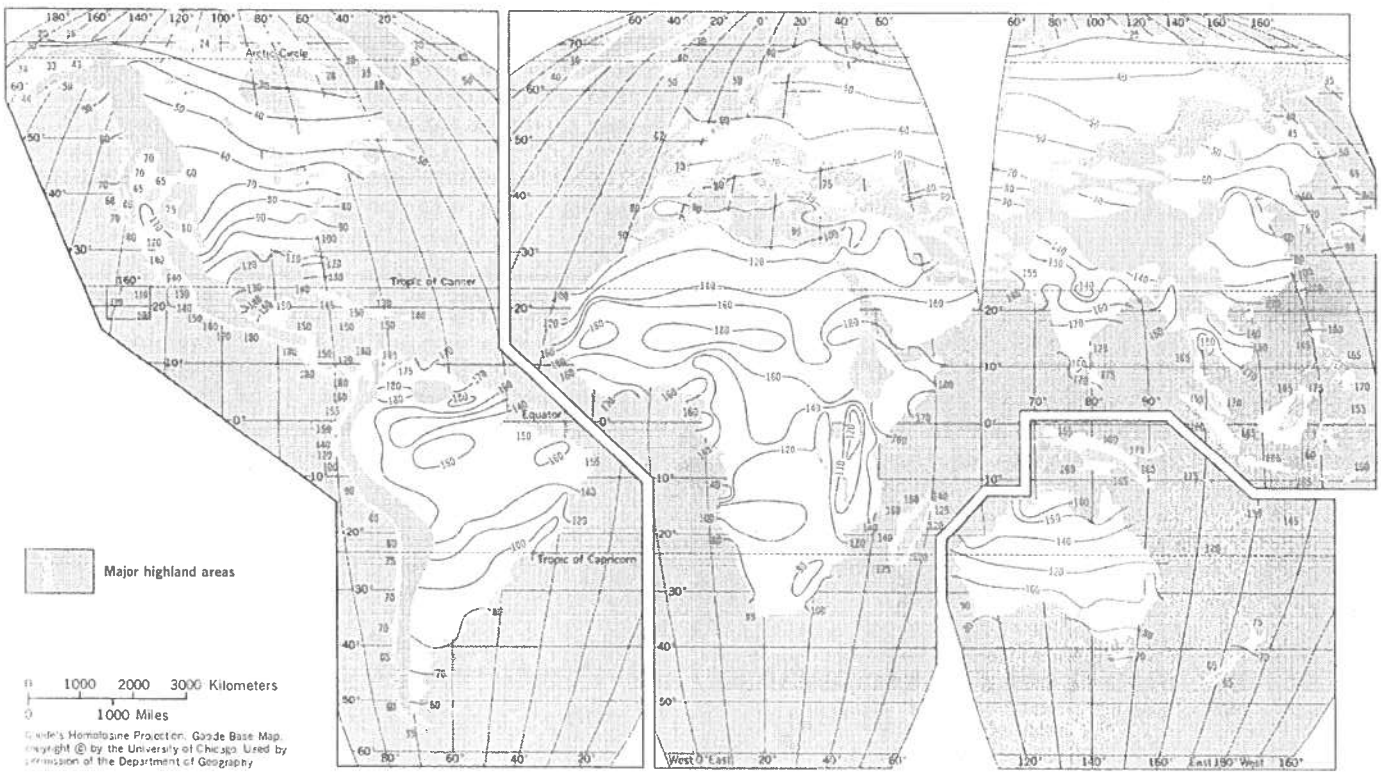
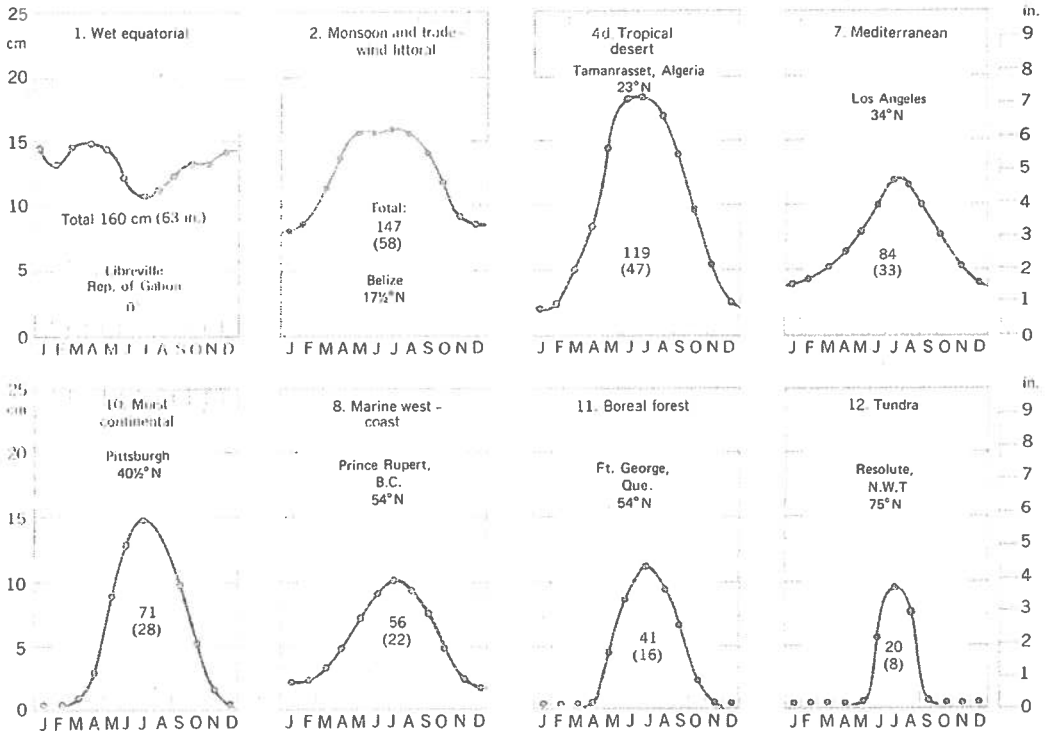


FIGURE 9.12 World map of annual water need (potential evapotranspiration). (Data of C. W. Thornthwaite Associates, Laboratory of Climatology, Centerton, N.J. Based on Goode Base Map.)



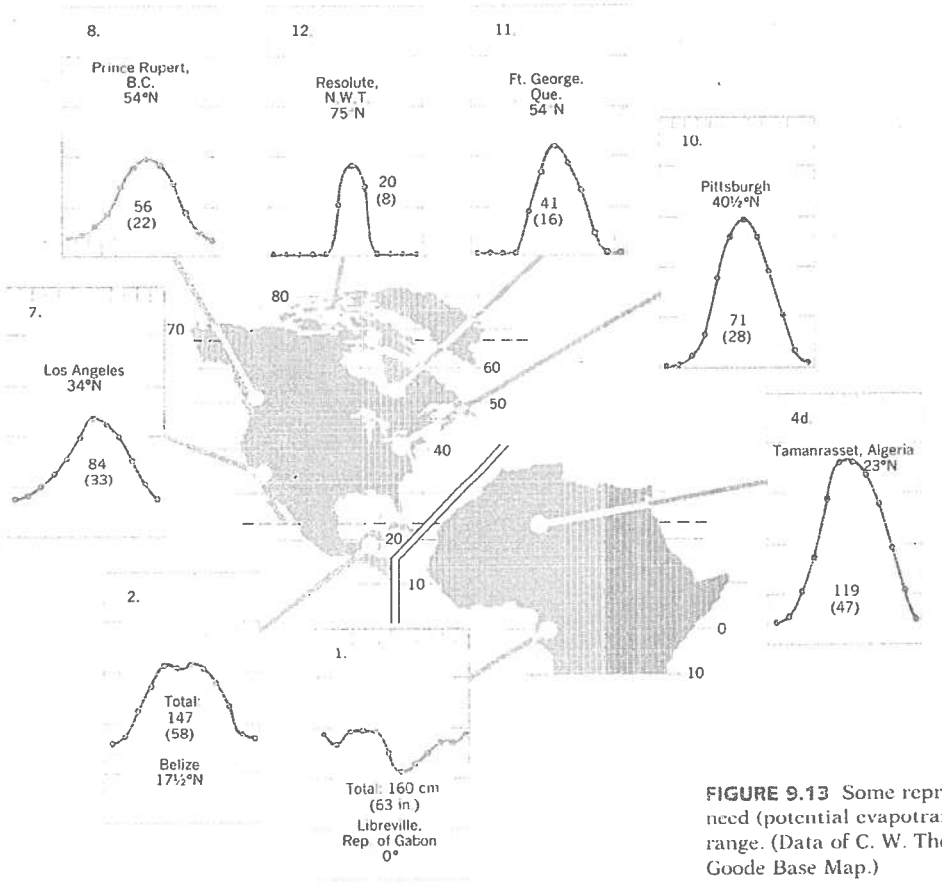


FIGURE 9.13 Some representative annual cycles of water need (potential evapotranspiration) over a wide latitude range. (Data of C. W. Thornthwaite Associates. Based on Goode Base Map.)

CHAPTER 16

RUNOFF, STREAMS, AND GROUND WATER

The primary concern of this chapter is with the science of *hydrology*, which is a study of water as a complex but unified system on the earth. We began our investigation of hydrology in Chapter 9 by covering that phase of hydrology in which soil water is recharged by precipitation and returned directly to the atmosphere by evapotranspiration, or to be disposed of as runoff.

Recall that many soil-water budgets show a substantial water surplus, to be disposed of as runoff. Our investigation now deals with that water surplus and the paths it follows as subsurface water and surface water. There are two basic paths of escape for surplus water. First, surplus water may flow over the ground surface as runoff to lower levels. As it travels, the dispersed flow becomes collected into streams, which eventually conduct the runoff to the ocean. Second, surplus water may percolate through the soil, traveling downward under the force of gravity to become part of the underlying ground-water body. Following subterranean flow paths, this water emerges to become surface water, or it may emerge directly in the shore zone of the ocean. In this chapter, we trace both the surface and subsurface pathways of flow of surplus water. In so doing, we will complete the hydrologic cycle.

RUNOFF AS A VITAL RESOURCE

Surplus fresh water, as runoff, is a vital part of the environment of terrestrial life forms and of humans in particular. Surface water in the form of streams, rivers, ponds, and lakes constitutes one of the distinctive environments of plants and animals.

Our heavily industrialized society requires enormous supplies of fresh water for its sustained operation. Urban dwellers consume water in their homes at rates of 150 to 400 liters (40 to 100 gallons) per person per day. Enormous quantities of water are used for cooling purposes in air-conditioning units and power plants.

In view of projections based on existing rates of increase in water demands, we shall be hard put in the future to develop the needed supplies of pure fresh water. Water pollution also tends to increase as populations grow and urbanization advances over broader areas. A disconcerting concept is that the available resource of pure fresh water is shrinking while demands are rising. Knowledge of hydrologic processes enables us to evaluate the total water resource, to plan for its management, and to protect it from pollution.

FORMS OF OVERLAND FLOW

Figure 16.1 shows what happens to rain falling on a hillslope that has its natural soil and plant cover intact. Some rain is caught and held above the ground on leaves and stems, a process called *interception*. Intercepted water may be returned directly to the atmosphere by evaporation. Precipitation reaching the soil surface enters the soil by infiltration; but when the rate of precipitation exceeds the rate of infiltration, water begins to accumulate as surface detention in minor surface depressions in the soil or behind small check dams of plant litter. The phenomenon is also called *depression storage*. Some of this water is evaporated directly into the atmosphere; the remainder eventually infiltrates the soil.

Runoff that flows down the slopes of the land in

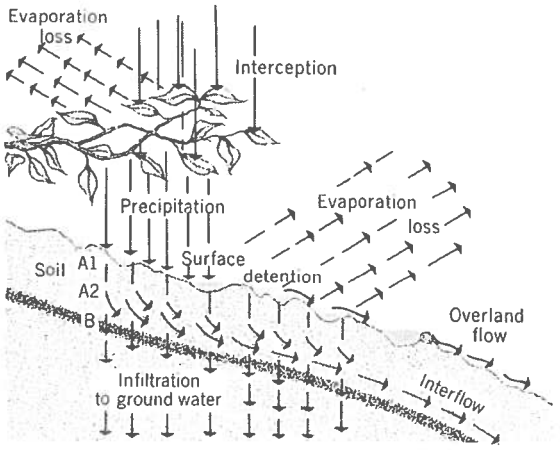


FIGURE 16.1 Precipitation, interception, and overland flow.

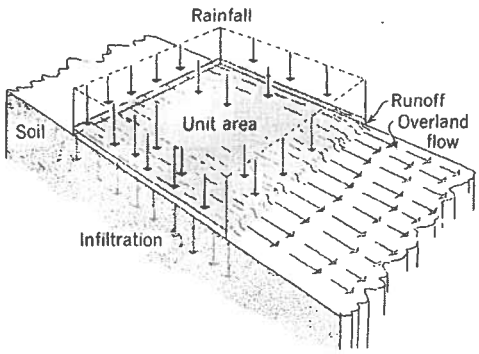


FIGURE 16.2 Rainfall, infiltration, and overland flow. (From A. N. Strahler, *The Earth Sciences*, 2nd ed., Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

broadly distributed sheets is referred to as *overland flow* (Figure 16.2). We distinguish overland flow from *stream flow (channel flow)*, in which the water occupies a narrow channel confined by lateral banks. Overland flow can take several forms. It may be a continuous thin film, called *sheet flow*, where the soil or rock surface is smooth. Flow may take the form of a series of tiny rivulets connecting one water-filled hollow with another where the ground is rough or pitted. On a grass-covered slope, overland flow is subdivided into countless tiny threads by water passing around the stems. Even in a heavy and prolonged rain, you might not notice overland flow in progress on a sloping lawn. On heavily forested slopes, overland flow may pass entirely concealed beneath a thick mat of decaying leaves. At the base of a hillslope, overland flow is disposed of by passing into the head of a gully, a stream channel, or a lake (Figure 16.3). In some cases, overland flow simply disappears into a porous layer of sandy soil.

Overland flow is measured in centimeters or inches of water per hour, just as for precipitation and infiltration. A simple formula expresses the rate at which overland flow will be produced by a given unit of ground surface as follows:

Rate of production of overland flow = rate of precipitation − rate of infiltration

For example, if the rate of infiltration became constant at a value of 1 cm/hr and the rate of rainfall was a steady 2 cm/hr (a heavy rain), the runoff would be produced at a rate of 1 cm/hr, assuming none to be returned to the atmosphere by evaporation.

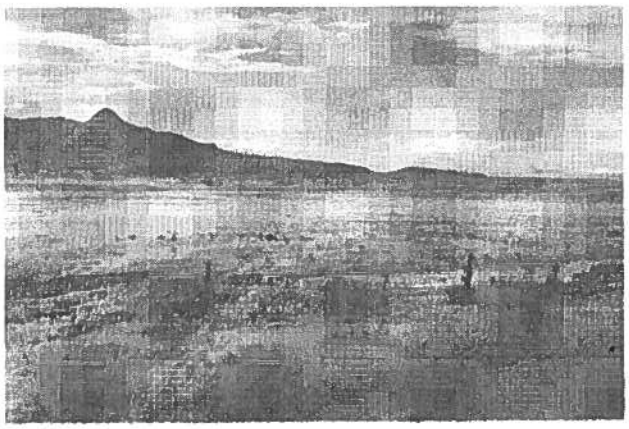


FIGURE 16.3 Overland flow taking the form of a thin sheet of water covers the nearly flat plain in the middle distance. This water is converging into stream flow in a narrow, steep-sided gully (left). The photograph was taken shortly after a summer thunderstorm had deluged the area. The locality, near Raton, New Mexico, shows steppe grassland vegetation. (Mark A. Melton.)

INTERFLOW

Mature soils of certain of the major soil orders have a dense layer (B horizon) of clay accumulation, but the soil layer above it (A horizon) is sandy and loose. (These horizons are explained in Chapters 23 and 24.) When the infiltrating water reaches the B horizon, its downward passage is greatly impeded. The soil water then backs up into the more permeable A horizon and begins to move in the downslope direction, parallel with the ground surface (Figure 16.1). This lateral flow is called *interflow* (or *throughflow*). Eventually interflow reaches the base of the hillslope and seeps into a stream channel. Interflow is thus a pathway of runoff intermediate in position and speed of travel between overland flow and deeper ground-water flow. Interflow is also important where a permeable soil rests on a dense, impermeable bedrock mass.

DRAINAGE SYSTEMS

Overland flow, interflow, and out-seeping ground-water flow eventually contribute to a stream, which is a much faster, more concentrated form of runoff (Figure 16.4). We define a *stream* as a long, narrow body of flowing water occupying a trenchlike depression, or channel, and moving to lower levels under the force of gravity.

The total system of downslope water flow from the point of arrival at the ground surface comprises the *drainage system*. It consists of a branched network of stream channels, as well as the sloping ground surfaces that contribute overland flow and interflow to those

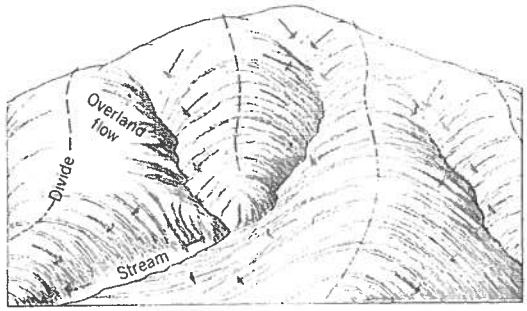


FIGURE 16.4 Overland flow from slopes in the headwater area of a stream system supplies water and rock debris to the smallest elements of the channel network.

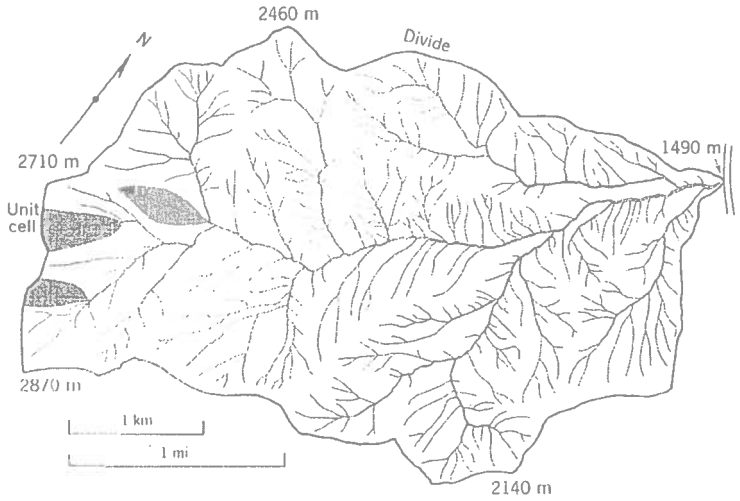


FIGURE 16.5 Channel network of the basin of Pole Canyon, Utah. (Data from U.S. Geological Survey and Mark A. Melton. From A. N. Strahler, *Planet Earth: Its Physical Systems Through Geologic Time*, Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

channels. The entire system is bounded by a drainage divide, outlining a more or less pear-shaped *drainage basin*. The ground slopes and channels are adjusted to dispose of, as efficiently as possible, the runoff and its contained load of mineral particles.

A typical stream network contributing to a single outlet is shown in Figure 16.5. Note that each fingertip tributary receives runoff from a small area of land surface surrounding the channel. This area may be regarded as the unit cell of the drainage system. The entire surface within the outer divide of the drainage basin constitutes the watershed for overland flow. A drainage system is a converging mechanism funneling and integrating the weaker and more diffuse forms of runoff into progressively deeper and more intense paths of activity.

STREAM CHANNEL GEOMETRY

A *stream channel* is a narrow trough, shaped by the forces of flowing water to be highly effective in moving the quantities of water and sediment supplied to the stream. Channels may be so narrow that a person can jump across them or as wide as 1.5 km (1 mi) for a great river—such as the Mississippi.

Hydraulic engineers who measure stream dimensions and flow rates have adopted a set of terms to describe channel geometry (Figure 16.6). Depth, d, in meters, is measured at any specified point in the stream as the vertical distance from surface to bed. Width, w, is the distance in meters across the stream from one water's edge to the other. The cross-sectional area, A, is the area in square meters of a vertical slice across the stream. The *wetted perimeter*, P, is the length of line of contact between the water and the channel, as measured from the cross section.

The rate of fall in altitude of the stream surface in the downstream direction is the *stream gradient;* it is often stated in meters per kilometer or feet per mile. A gradient of 5 m/km means that the stream surface undergoes a vertical drop of 5 m for each km of horizontal distance downstream. Slope can also be given in terms of percent grade, a common practice in engineering. A grade of 3 percent, or 0.03, means that the stream drops 3 m for every 100 m of horizontal distance.

STREAM FLOW

As a stream flows under the influence of gravity, the water encounters resistance—a form of friction—with the channel walls. As a result, water close to the bed and banks moves slowly; that in the deepest and most centrally located zone flows fastest. Figure 16.6 indicates by arrows the speed of flow at various points in the stream. The single line of maximum velocity is located in midstream, where the channel is straight and symmetrical.

Our statement about velocity needs to be qualified. Actually in all but the most sluggish streams, the water

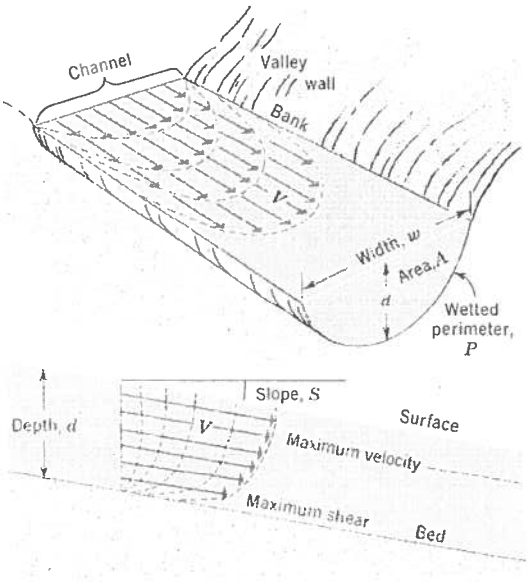


FIGURE 16.6 Stream flow within a channel is most rapid near the center and just below the water surface.

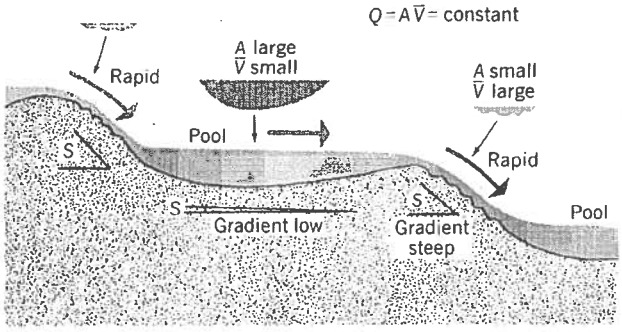


FIGURE 16.7 Schematic diagram of the relationship among cross-sectional area, mean velocity, and gradient. (From A. N. Strahler, *The Earth Sciences*, 2nd ed., Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

is affected by *turbulence*, a system of innumerable eddies that are continually forming and dissolving. A particular molecule of water, if we could keep track of it, would describe a highly irregular, corkscrew path as it is swept downstream. Motions include upward, downward, and sideward directions. Turbulence in streams is extremely important because of the upward elements of flow that lift and support fine particles of sediment. The murky, turbid appearance of streams in flood is ample evidence of turbulence, without which sediment would remain near the bed. Only if we measure the water velocity at a certain fixed point for a long period of time, say several minutes, will the average motion at the point be downstream and in a line parallel with the surface and bed. Average values are shown by the arrows in Figure 16.6.

Because the velocity at a given point in a stream differs greatly according to whether it is being measured close to the banks and bed or out in the middle line, a single value, the *mean velocity*, is needed. Mean velocity is computed for the entire cross section to express the activity of the stream as a whole.

STREAM DISCHARGE

A most important measure of stream flow is *discharge*, *Q*, defined as the volume of water passing through a given cross section of the stream in a given unit of time. Discharge is stated in cubic meters per second (abbreviated to cms); in English units, cubic feet per second (cfs). Discharge may be obtained by taking the mean velocity, *V*, and multiplying it by the cross-sectional area, *A*. This relationship is stated by the important equation, $Q = AV$.

We realize that water will flow faster in a channel of steep gradient than one of gentle gradient because the component of gravity acting parallel with the bed is stronger for the steeper gradient. As shown in Figure 16.7, velocity increases quickly where a stream passes from a wide pool of low gradient to a steep stretch of rapids. As V increases, A must decrease; otherwise their product, AV, would not be held constant. In the pool, where velocity is low, A is correspondingly increased.

Stream channels differ in the amount of frictional resistance that the bed and banks offer to the flow of water. Resistance is large in a broad but shallow channel and is much less in a deep, narrow channel. The optimum channel would be semicircular in cross section. Streams are, however, usually required to have broad, shallow channels because of the load of mineral particles that must be carried or because the banks are weak and will not hold a steep attitude.

STREAM GAUGING

Information on daily discharges of streams is vital to a nation, not only as a measure of the surface-water resource, but also for the design of flood-protection structures and for the prediction of stream floods as they progress down a river system.

In the United States, the measurement of stream flow, or *stream gauging*, is under the jurisdiction of the U.S. Geological Survey. In cooperation with states and municipalities, that organization maintains over 11,000 gauging stations on principal streams and their tributaries. Figures on discharge are published by the Geological Survey in a series of Water-Supply Papers.

A stream-gauging station requires a device for measuring the height of the water surface, or *stream stage*. Simplest to install is a *staff gauge*, which is simply a graduated vertical scale attached to a post or bridge pier. This must be read directly by an observer whenever the stage is to be recorded. More useful is an automatic-recording gauge, which is mounted in a stilling tower built beside the river bank (Figure 16.8).

To measure stream discharge, one must determine both the area of cross section of the stream and the mean velocity. Velocity measurement requires a *current meter*. The meter is lowered into the stream at closely spaced intervals so the velocity can be read at a large number of points evenly distributed in a grid pattern through the

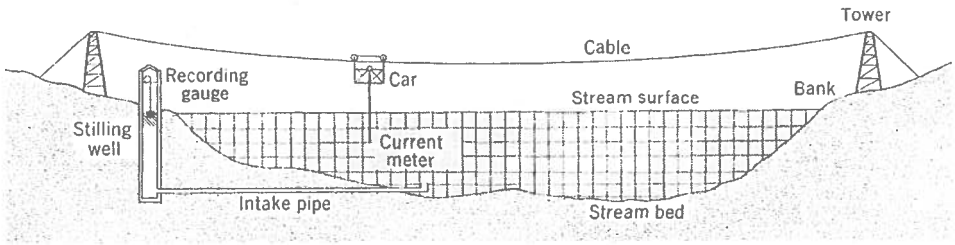


FIGURE 16.8 Idealized diagram of stream-gauging installation

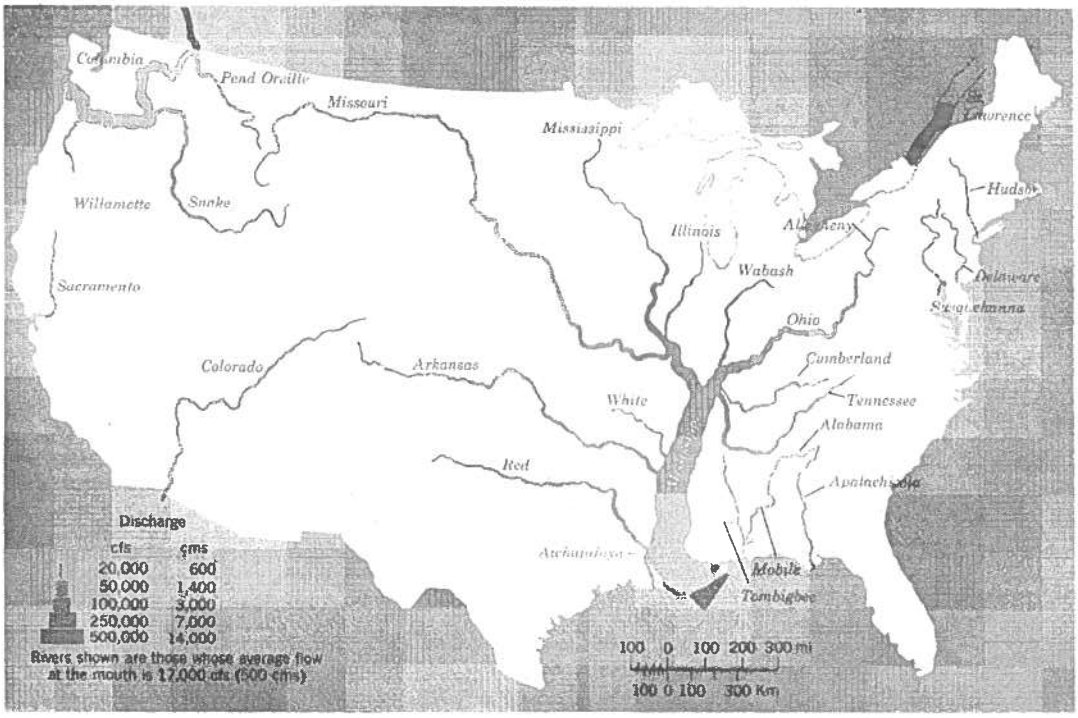


FIGURE 16.9 A flow map of the United States showing the relative magnitude of rivers by means of a broad line with width proportional to the mean annual water discharge. (U.S. Geological Survey.)

stream cross section (Figure 16.8). A bridge often serves as a convenient means of crossing over the stream; otherwise, a cable car or small boat is used. Revolving cups on the current meter turn at a rate proportional to current velocity. As the velocities are being measured from point to point, a profile of the riverbed is also being made by sounding the depth. From these readings, a profile is drawn and the cross-sectional area is measured from the profile. Mean velocity is computed by summing all individual velocity readings and dividing by the number of readings. Discharge can then be computed using the formula $Q = AV$.

Figure 16.9 is a map showing the relative discharge of major rivers of the United States. ("River" is a popular term applied to a large stream. The word "stream" is the scientific term designating channel flow of any magnitude of discharge.) The mighty Mississippi with its tributaries dwarfs all other North American rivers, although the MacKenzie, Columbia, and Yukon rivers, as well as the Great Lakes discharge through the St. Lawrence River, are also of major proportions. The Colorado River, a much smaller stream, crosses a vast semiarid and arid region in which little tributary flow is added to the snowmelt source high in the Rocky Mountains.

STREAM FLOW AND PRECIPITATION

It seems obvious that the discharge of a stream will increase in response to a period of heavy rainfall or snowmelt. The response is delayed, of course, but the length of delay depends on a number of factors. The most important factor is the size of the drainage basin feeding the stream above the place where the gauge is located. The relationship between stream discharge and precipitation is best studied by means of a simple graph, called a *hydrograph.*

Figure 16.10 is a hydrograph for a drainage basin about 800 sq km (300 sq mi) in area, located in Ohio in a region of moist continental climate. The graph gives data for a 2-day summer storm. Rainfall is shown by a bar graph giving the precipitation (cm) in each 2-hour period. Also plotted on the graph (smooth line) is the discharge (cms) of Sugar Creek, the trunk stream of the drainage basin. The average total rainfall over the watershed of Sugar Creek was about 15 cm (6 in.); of this amount half passed down the stream within 3 days' time. Some rainfall was held in the soil as soil water, some evaporated, and some infiltrated to the water table to be held in long-term storage in the ground-water body.

Studying the rainfall and runoff graphs in Figure 16.10, we see that prior to the onset of the storm, Sugar Creek was carrying a small discharge. This flow, being supplied by the slow seepage of ground water into the channel, is termed *base flow.* After the heavy rainfall began, several hours elapsed before the stream gauge at the basin mouth began to show a rise in discharge. This interval, called the *lag time,* indicates that the branching system of channels was acting as a temporary reservoir. As the stage was rising, water was soaking into permeable bank materials, where it was being temporarily stored.

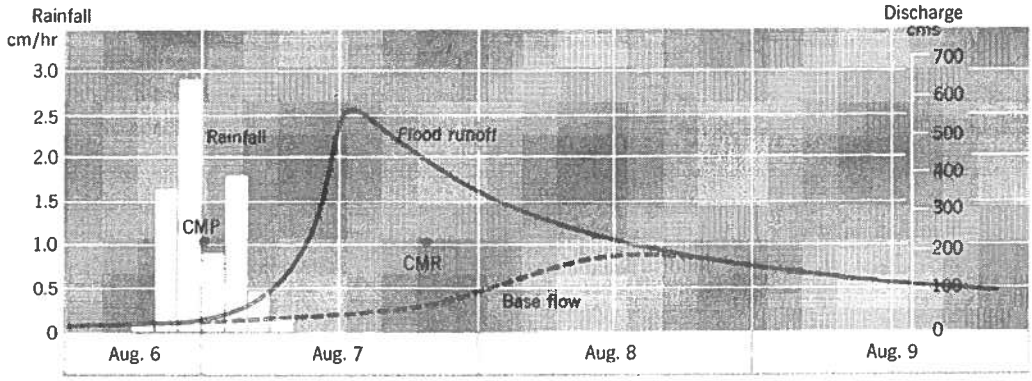


FIGURE 16.10 Four days of flow of Sugar Creek, Ohio. (After William G. Hoyt and Walter B. Langbein, *Floods;* copyright 1955 by Princeton University Press; Figs. 8 and 13, pp. 39 and 45. Reprinted by permission of Princeton University Press.)

Lag time is measured as the difference between center of mass of precipitation (CMP) and center of mass of runoff (CMR), as labeled in Figure 16.10. The peak flow of Sugar Creek was reached almost 24 hours after the rain began; the lag time was about 18 hours. Note also that the rate of decline in discharge was much slower than the rate of rise because overland flow was followed by contributions from interflow and, finally, from ground-water seepage.

In general, the larger a watershed, the longer is the lag time between peak rainfall and peak discharge and the more gradual is the rate of decline of discharge after the peak has passed. Notice that the flow of Sugar Creek showed a slow but distinct rise in the amount of discharge contributed by base flow.

BASE FLOW AND OVERLAND FLOW

In regions of moist climates, where the water table is high and normally intersects the important stream channels, the hydrographs of larger streams will show clearly the effects of three sources of water: overland flow, interflow, and base flow. Figure 16.11 is a hydrograph of the Chattahoochee River, Georgia, a large stream draining a watershed of some 8700 sq km (3350 sq mi), much of it in the moist southern Appalachian Mountains. The sharp, abrupt fluctuations in discharge are produced by overland flow and interflow following rain periods of 1 to 3 days' duration. These are each similar to the hydrograph of Figure 16.10, except that they are shown in Figure 16.11 much compressed by the time scale.

After each rain period, the discharge falls off rapidly; but, if another storm occurs within a few days, the discharge rises to another peak. The enlarged inset graph shows details for the month of January. When a long period intervenes between storms, the discharge falls to a low value, the base flow, where it levels off.

Throughout the year the base flow, which represents ground-water inflow into the stream, undergoes a marked annual cycle. During the period of recharge (winter and early spring), water-table levels are raised and the rate of inflow into streams is increased. For the Chattahoochee River, the rate of base flow during January, February, March, and April holds uniform at about 110 cms (4000 cfs). The base flow begins to decline in spring,

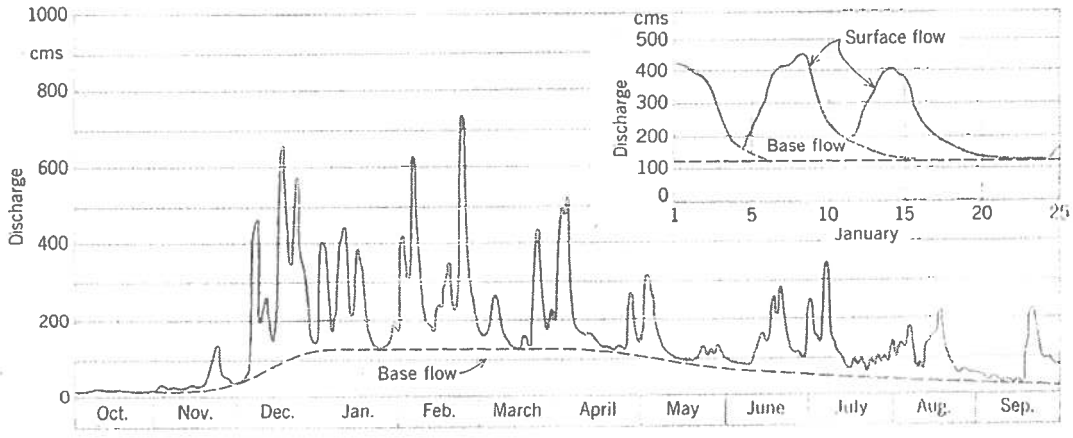


FIGURE 16.11 Flow peaks of the Chattahoochee River. (Data from U.S. Geological Survey in E. E. Foster, *Rainfall and Runoff.*)

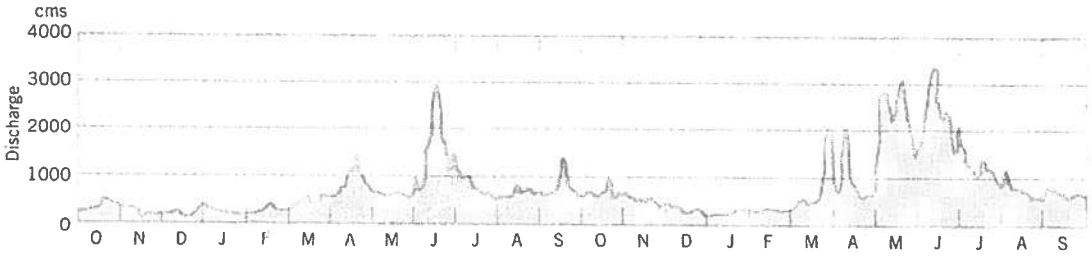


FIGURE 16.12 Discharge of the Missouri River. (Data from U.S. Geological Survey in E. E. Foster, *Rainfall and Runoff*.)

when heavy evapotranspiration losses reduce soil water and therefore cut off the recharge of ground water. The decline continues throughout the summer, reaching a low of about 30 cms (1000 cfs) by the end of October.

Finally, examine the hydrograph of the Missouri River at Omaha, Nebraska, for a 2-year period of record starting in October (Figure 16.12). This great river, draining 840,000 sq km (322,800 sq mi) of watershed, is a major tributary of the Mississippi River. Note that the discharge, ranging from 280 to 2800 cms (10,000 to 100,000 cfs), is many times greater than the discharges of smaller streams considered thus far. High rates of flow are chiefly from snowmelt, which occurs on the High Plains in spring and in the Rocky Mountain headwater areas in early summer. This event explains the sudden high discharges from April through June. During midwinter, when soil water is frozen and total precipitation is small over the watershed as a whole, the discharge rises little above the base flow. Ground-water recharge occurring in the spring raises summer levels of base flow to about 570 cms (20,000 cfs), or 2 to 3 times the winter base flow.

FLOODS

Everyone has seen enough news-media photos of river floods to have a good idea of the appearance of floodwaters and the havoc wrought by their erosive power and by the silt and clay they leave behind. Even so, it is not easy to define the term *flood*. Perhaps it is enough to say that a condition of flood exists when the discharge of a river cannot be accommodated within the margins of its normal channel, so the water spreads over adjoining ground on which crops or forests are able to flourish.

Most larger streams of moist climates have a *floodplain*, a belt of low, flat ground bordering the channel on one or both sides inundated by stream waters about once a year. This flood usually occurs in the season when abundant supplies of surface water combine with the effects of a high water table to supply more runoff than can stay within the channel. Such an annual inundation is considered a flood, even though its occurrence is expected and does not prevent the cultivation of crops after the flood has subsided. The seasonal inundation does not interfere with the growth of dense forests, which are widely distributed over low, marshy floodplains in all moist regions of the world. Still higher discharges of water, the rare and disastrous floods that may occur as infrequently as 3 to 5 decades, inundate ground lying well above the floodplain (Figure 16.13).

For practical purposes, the National Weather Service, which provides a flood-warning service, designates a particular stage of gauge height at a given place as the *flood stage;* this implies that the critical level has been reached above which *overbank flooding* may be expected to set in. Immediately at or below flood stage, the river may be described as being in the *bank-full stage*, the flow being entirely within the limits of the heavily scoured channel.

DOWNSTREAM PROGRESS OF A FLOOD WAVE

The rise of a river stage to its maximum height, or *flood crest*, followed by a gradual lowering of stage, is termed the *flood wave*. The flood wave is simply a large-sized rise and fall of river discharge of the type already analyzed in earlier paragraphs, and it follows the same principles. Figure 16.14*A* shows the downstream progress of a flood on the Chattooga-Savannah river system. On the Chattooga River near Clayton, Georgia, the flood peak, or crest, was reached quickly—1 day after the storm—and subsided quickly. On the Savannah River, 105 km (65



FIGURE 16.13 The city of Harrisburg, Pennsylvania, lies partly submerged beneath flood waters generated by Hurricane Agnes, June 1972. The Susquehanna River, seen in the distance, rose to nearly 5 m (16 ft) above flood stage, inundating the low river terrace on which the downtown portion of Harrisburg is built. (Department of the Army.)

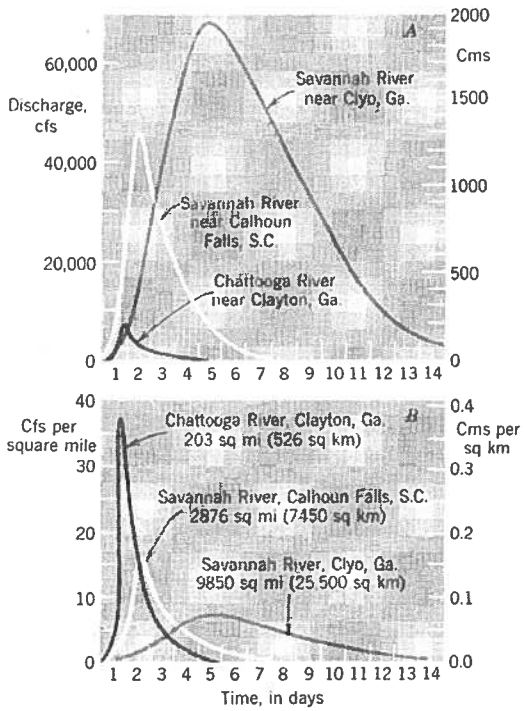


FIGURE 16.14 The downstream progress of a flood wave. (After William G. Hoyt and Walter B. Langbein, *Floods;* copyright 1955 by Princeton University Press; Figs. 8 and 13, pp. 39 and 45. Reprinted by permission of Princeton University Press.)

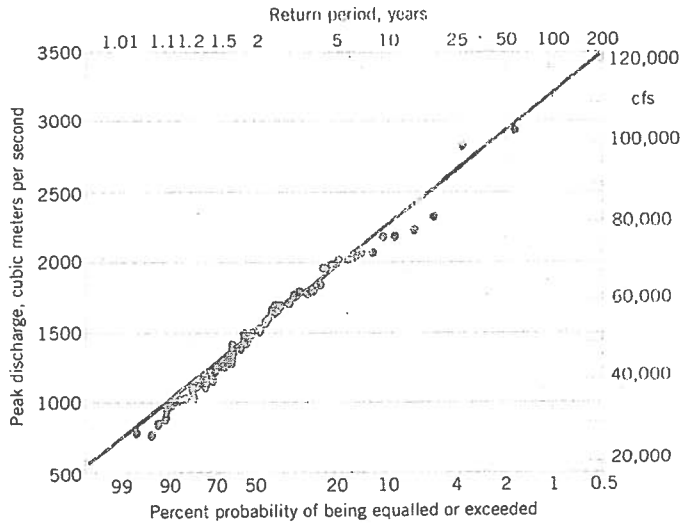


FIGURE 16.15 Flood frequency data for the Clearwater River at Kamaiah, Idaho. Each dot is a measured maximum yearly discharge in a 53-year record. (Data of U.S. Geological Survey from R. K. Linsley, M. A. Kohler, and J. L. Paulhus, *Hydrology for Engineers*, second edition, McGraw-Hill, New York, p. 347.)

mi) downstream at Calhoun Falls, South Carolina, the peak flow occurred a day later; but the discharge was much larger because of the larger area of watershed involved. Downstream another 153 km (95 mi), near Clyo, Georgia, the Savannah River crested 5 days after the initial storm, with a discharge of over 1700 cms (60,000 cfs). This set of three hydrographs shows that (1) the lag time in occurrence of the crest increases downstream; (2) the entire period of rise and fall of flood wave becomes longer downstream; and (3) the discharge increases greatly downstream as watershed area increases.

Figure 16.14*B* is a somewhat different presentation of the same flood data in that the discharge is given in terms of a common unit of area (cms/sq km), thus eliminating the effect of increase in discharge downstream and showing us only the shape, or form, of the flood crest.

FLOOD PREDICTION

The National Weather Service operates a River and Flood Forecasting Service through 85 offices located at strategic points along major river systems of the United States. Each office issues river and flood forecasts to the communities within the associated district, which is laid out to cover one or more large watersheds. Flood warnings are publicized by every possible means. Close cooperation is maintained with various agencies to plan the evacuation of threatened areas and the removal or protection of vulnerable property.

Hydrologists use sophisticated statistical methods to arrive at estimates of the probability that a flood of a given stage of discharge will occur in any given year. Estimates are, of course, based on gauge records of past years. Figure 16.15 is a graph on which the greatest discharge of each year of record is plotted as a dot. Discharge in units of cms is plotted on the vertical axis in equally spaced units (arithmetic scale). Numbers on the bottom horizontal scale tell the probability that the given discharge will be equaled or exceeded in a given year. For example, a discharge associated with the value of 20 percent is interpreted to mean "a discharge of this magnitude (2000 cms) can be expected to be equaled or exceeded in 20 out of 100 years." The numbers on the scale at the top of the graph tell the return period (or recurrence interval), which is simply the inverse value of the probability percentage on the bottom scale. In the example just cited, probability 20 percent, the return period is 5 years, meaning that on the average the stated discharge (2000 cms) can be expected to be equaled or exceeded once every 5 years. The horizontal scale on the graph has been adjusted to obtain the best possible fit of the plotted points to a straight line. The fitted line can be expressed by a mathematical equation; it is one of several estimating equations that might be applied to the same data in efforts to secure the best predictions.

A much simpler pictorial analysis is used in Figure 16.16 to suggest the likelihood that a given maximum stage will occur in a given calendar month of the year. Each bar divides all observed maximum stages of the calendar month into quartiles, or groups of 25 percent of the occurrences. The dots at top and bottom of the bar give the absolute maximum and minimum stages observed during the entire period of record.

The graph for the Mississippi River at Vicksburg illustrates a great river responding largely to spring floods

so as to yield a simple annual cycle. All floods have occurred in the first 6 months of the year. The Colorado River at Austin, Texas, illustrates a river draining largely semiarid plains. Summer floods are produced directly by torrential rains from invading moist tropical air masses. Floods of the late summer and fall are often attributable to tropical storms (hurricanes) moving inland from the Gulf of Mexico. The Sacramento River at Red Bluff, California, has a winter flood season when rains are heavy, but a sharp dip to low stages in late summer, which is the very dry period in the Mediterranean climate. The flood expectancy graph for the Connecticut River at Hartford shows two seasons of floods. The more reliable of the two is the early spring, when snowmelt is rapid over the mountainous New England terrain. The second season is in the fall, when rare but heavy rainstorms, some of hurricane origin, bring exceptional high stages.

The regulation of floods of large rivers is described in Chapter 17 in connection with landforms of floodplains.

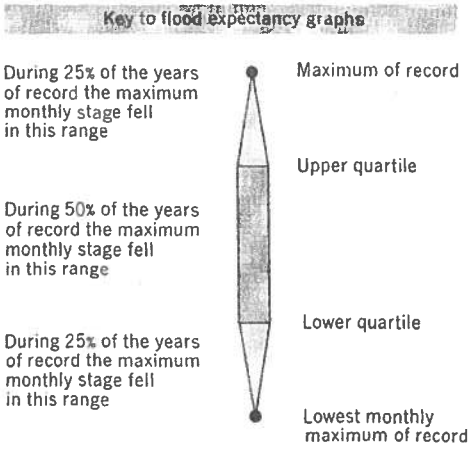


FIGURE 16.16 (Above and right) The highest stage that occurred in each month is given in terms of percentages on these graphs of four rivers. (National Weather Service.)

FLASH FLOODS AND THEIR PREDICTION

Known as *flash floods*, sudden and short-lived overbank discharges differ in important respects from the prolonged floods of large streams we have thus far described. Flash floods quickly follow torrential thunderstorm rains that deliver on the order of 20 to 30 cm (8 to 12 in.) in a 24–hour period over a small hilly or mountainous watershed within which ground slopes and stream channel gradients are steep. Three weather situations can be recognized as producing flash floods. One is a hurricane that has penetrated the continental interior; Camille, Agnes, and Amelia produced destructive flash floods. Another, typical of the central and eastern United States, is a slow-moving or stagnant weather front along which large amounts of moist tropical air are being imported. A third, found in the mountainous western states, is orographic in nature, typified by flash floods developed over California mountain ranges in the winter months.

Because of the small size of the affected watershed, the lag time of a flash flood is extremely short—perhaps only an hour or two—giving little or no time to warn inhabitants of the mainstream environs. The flood arrives as a swiftly moving wall of turbulent water, sweeping away buildings, vehicles, and retaining walls in its path. In heavily forested watersheds, as in the Appalachians, the water carries little bedload and is highly erosive. In arid western watersheds, great quantities of coarse rock debris are swept into the main channel and travel with the floodwater. An example is the Big Thompson River flash flood of July 1976, occurring in a watershed of the Colorado Front Range. (Channel erosion from this flood is pictured in Figure 17.8.) In Chapter 18, we will refer to flash floods in the desert (see Figure 18.8).

An additional hazard of the flash flood is the breaching of an earthfill dam across the main channel, suddenly releasing an enormous quantity of stored water. Just such an occurrence accompanied the great Johnstown Flood of May 1889, in the Conemaugh Valley of the Allegheny Plateau, in western Pennsylvania; it drowned more than 2100 persons out of a town population of 30,000. A similar flash flood in 1977 hit the same valley, taking 76 lives—this in spite of extensive flood control measures that had been implemented between the two floods.

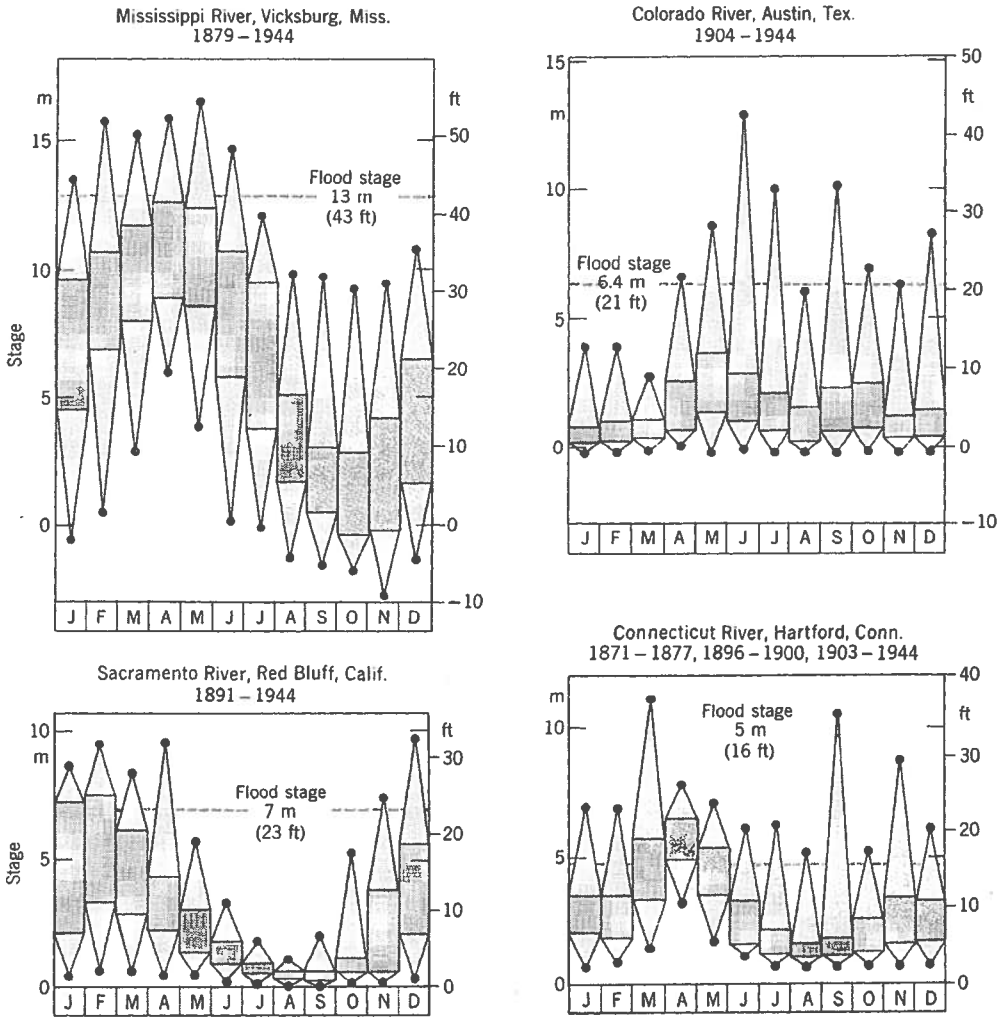
The National Weather Service operates flash flood warning systems at some 55 locations throughout the United States. Included in the instrumentation of each key observing point is an electronic river level sensor which immediately signals to an alarm station that a critical water level for flash flooding has been reached. The alarm station releases various forms of warning devices, such as sirens and lights. Quick action is of the essence, because many flash floods are generated during night hours and lag time is so very brief. In recent decades, about 200 lives have been lost annually in floods, and flash floods accounted for 80% of those deaths.

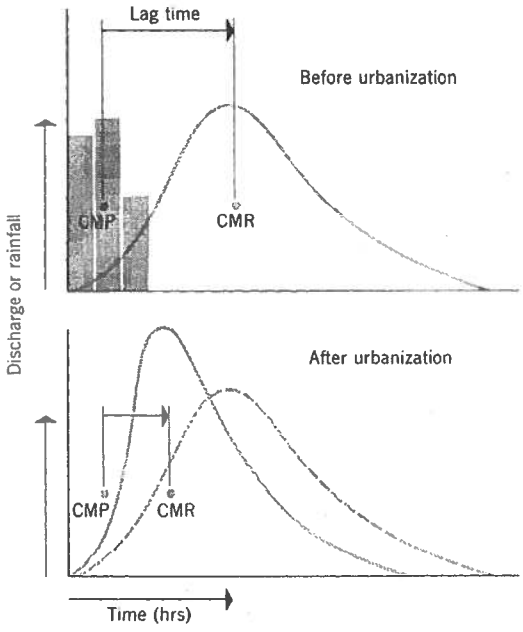
HYDROLOGIC EFFECTS OF URBANIZATION

Hydrology of a watershed is altered in two ways by urbanization. First, an increasing percentage of the surface is rendered impervious to infiltration by the construction of roofs, driveways, walks, pavements, and parking lots. It has been estimated that in residential areas, for a lot size of 1400 sq m (15,000 sq ft), the impervious area amounts to about 25 percent; for a lot size of 560 sq m (6000 sq ft), the impervious area amounts to about 80 percent.

An increase in proportion of impervious surface reduces infiltration and increases overland flow from the urbanized area. An important result is to increase the frequency and height of flood peaks during heavy storms. There is also a reduction of recharge to the ground-water body beneath; this reduction, in turn, decreases the base-flow contribution to channels in the area. Thus the full range of stream discharges, from low stages in dry periods to flood stages, is made greater by urbanization.

A second change caused by urbanization is the introduction of storm sewers that allow storm runoff from





paved areas to be taken directly to stream channels for discharge. Runoff travel time to channels is shortened at the same time that the proportion of runoff is increased by expansion in impervious surfaces. The two changes together conspire to reduce the lag time, as shown by the schematic hydrographs in Figure 16.17.

Many rapidly expanding suburban communities are soon finding that certain low-lying residential areas, formerly free of flooding, are being subjected to inundation by overbank flooding of a nearby stream. The need for careful terrain study and land-use planning is obvious in such cases to protect the unwary home buyer from locating in a flood-prone neighborhood.

One partial solution to the problem created by storm sewering is to return water runoff to the ground-water body by means of infiltrating basins. This program has

FIGURE 16.17 (left) Schematic hydrographs showing the effect of urbanization on lag time and peak discharge. Points CMP and CMR are centers of mass of rainfall and runoff, respectively, as in Figure 16.10. (After L. B. Leopold, U.S. Geological Survey, Circular 554.)

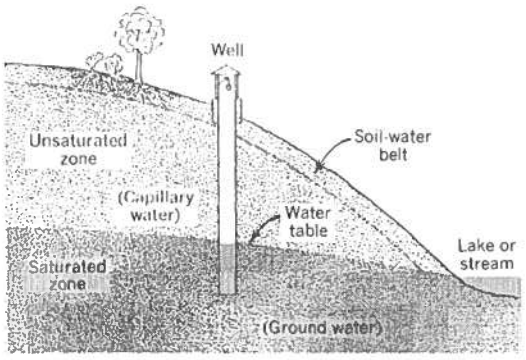


FIGURE 16.18 Zones of subsurface water.

been adopted on Long Island, in New York, where infiltration rates are high in sandy glacial materials. Another method of disposal of storm runoff is by recharge wells. At Orlando, Florida, storm runoff enters wells that penetrate cavernous limestone. The capacity of the system to absorb runoff without clogging appears to be adequate. In Fresno, California, a large number of gravel-packed wells of 76-cm (30-in.) diameter receive runoff from streets. The system has proved successful in disposing of storm drainage.

GROUND WATER

Ground water is that part of the subsurface water that fully saturates the pore spaces of the rock mass or soil. The ground water occupies the *saturated zone* (Figure 16.18). Above it is the *unsaturated zone,* in which water does not fully saturate the pores. The upper surface of the saturated zone is the *water table.* Water is held in the unsaturated zone by capillary force in tiny films adhering to the mineral surfaces. The unsaturated zone may be absent or very shallow where the saturated zone is at or close to the surface in low, flat regions.

At the base of the unsaturated zone is the *capillary fringe,* a thin layer in which the water has been drawn upward from the ground-water table through capillary force. The action is much like the rise of kerosene in a lamp wick or of water in a blotter. Water in the capillary fringe largely fills the soil pores and is thus continuous with the ground-water body. Thickness of the capillary fringe depends on the soil texture because capillary rise is higher when the openings are smaller. In a silty material, the capillary fringe may be 1 m (3 ft) thick, but as little as 1 cm thick in coarse sand or fine gravel with large pore spaces. (See Table 12.2.)

The true position of the water table is shown by the level of standing water in a well drilled or dug to some depth below the water table. Where wells are numerous in an area, the position of the water table can be mapped in detail by plotting the water heights and noting the trends in elevation from one well to the other. The water table is highest under the highest areas of surface—hilltops and divides—but descends toward the valleys, where it appears at the surface close to streams, lakes, or marshes (Figure 16.19). The reason for such a configuration of the water table is that water percolating down through the unsaturated zone tends to raise the water table, whereas seepage into streams, swamps, and lakes tends to draw off ground water and lower its level.

GROUND-WATER MOVEMENT

Because ground water moves extremely slowly, a difference in water-table level, or *hydraulic head,* is build up and maintained between areas of high elevation and those of low elevation. In periods of water surplus accompanying abnormally high precipitation, this head is increased by a rise in the water-table elevation under divide areas; in periods of water deficit, occasioned by drought, the water table falls (Figure 16.19).

In humid climates having a strong seasonal cycle of precipitation, and in climates in which soil water is frozen for several months of the year, a seasonal cycle of rise and fall of the water table is produced. This cycle is illustrated by a graph of water-table fluctuations in an observation well on Cape Cod, Massachusetts (Figure 16.20). Percolating water reaches the water table in abundant quantities from late winter to late spring, a phenomenon known as *ground-water recharge.* This recharge period, which reflects the period of water surplus in the soil–water budget, causes a seasonal rise in the water table. Recharge declines to very small amounts from midsummer to early winter, and during this time the water table steadily declines as water moves to lower levels under the force of gravity. The effect of a major drought shows in the graph by a downward trend of the average elevation in 1965 and 1966.

The subsurface phase of the hydrologic cycle is completed when the ground water emerges in places where the water table intersects the ground surface. Such places are the channels of streams and the floors of marshes and lakes. By slow seepage and spring flow, the water enters fast enough to balance the rate at which water enters the ground-water table by percolation from above.

Figure 16.21 shows paths of flow of ground water. Flow takes paths curved concavely upward. Water enter-

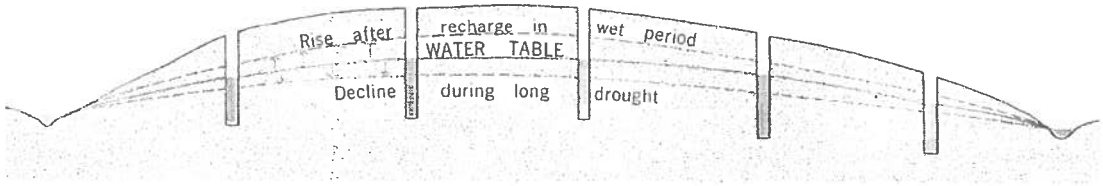


FIGURE 16.19 The water table conforms roughly to surface aphy.

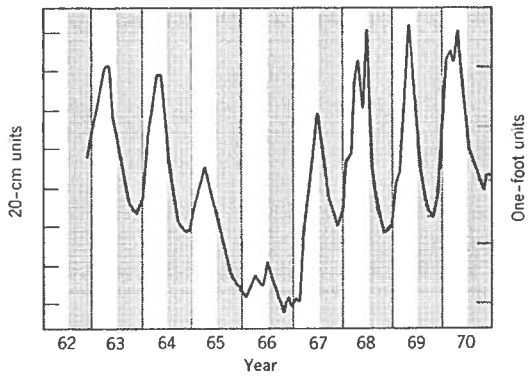


FIGURE 16.20 Hydrograph of an observation well on Cape Cod, Massachusetts, showing the characteristic annual cycle of rise and fall of the water table. (Data from U.S. Geological Survey.)

ing the hillside midway between divide and stream flows rather directly. Close to the divide point on the water table, however, the flow lines go almost straight down to great depths, from which they recurve upward to points under the streams. Progress along these deep paths is incredibly slow; that near the surface is much faster. The most rapid flow is close to the place of discharge in the stream, where the arrows are shown to converge.

GROUND WATER IN STRATA

Sedimentary strata often exert a strong control over the storage and movement of ground water. Clean, well-sorted sand—such as that found in beaches, dunes, or stream alluvium—can hold ground water in an amount equal to about one-third its bulk volume. The *porosity* of this material is high. Ground water moves freely through such coarse sediments, whose *permeability* is high. Coarse sandstone strata may retain relatively high porosity and permeability, becoming *aquifers* that both store ground water in substantial quantities and permit it to move freely within the strata. Other stratified permeable materials that form good aquifers are volcanic tephra and some kinds of frothy (scoriaceous) lavas. Limestone that has been rendered cavernous by the action of carbonic acid may also form an excellent aquifer. In contrast, beds of clay and shale have low permeability and resist the movement of ground water; they are known as *aquicludes*. Dense lava flows and sills may also serve as aquicludes.

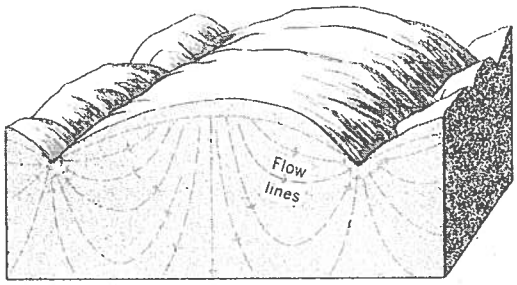


FIGURE 16.21 Theoretical paths of ground-water movement under divides and valleys. (Data from M. K. Hubbert. Drawn by A. N. Strahler.)

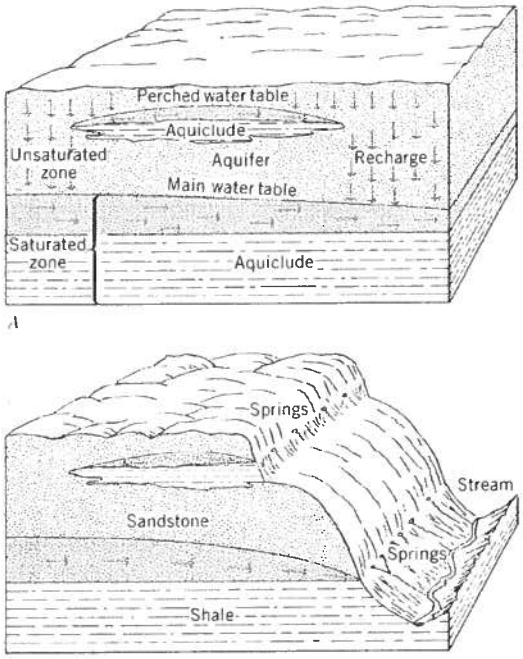


FIGURE 16.22 Ground water in horizontal strata. (*A*) An unconfined aquifer with an overlying aquiclude and a perched water table. (*B*) Springs controlled by the upper surfaces of aquicludes. (From A. N. Strahler, *Physical Geology*, Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

Where strata lie nearly horizontal or are gently dipping, the ground-water flow paths may be quite different from the simple patterns shown in Figure 16.21. Suppose for example, that the region is one of sedimentary strata with beds of sandstone alternating with beds of shale (Figure 16.22*A*). The main water table is shown to lie within the sandstone aquifer. The upper portion of the saturated zone occupies the lower part of the aquifer. The shale aquiclude is also in the saturated zone, but because of low permeability the water it holds is, for all practical purposes, stagnant. Ground water moves rather rapidly through the aquifer. The direction of this motion must necessarily be almost horizontal and is in the direction of downward slope (gradient) of the water table.

Figure 16.22 also shows a lenslike body of shale within the upper part of the sandstone formation. Acting as an aquiclude, this lens blocks the downward percolation of water through the unsaturated zone. As a result, a lenslike body of ground water collects above the shale lens. The upper surface of this ground-water lens is known as a *perched water table*. Ground water in this lens moves toward the edges of the lens, from which it trickles down through the lower part of the aquifer and eventually reaches the main water table.

The aquifer shown in Figure 16.22 is described as an *unconfined aquifer*, because the water table is free to receive recharge from above and can rise or fall freely

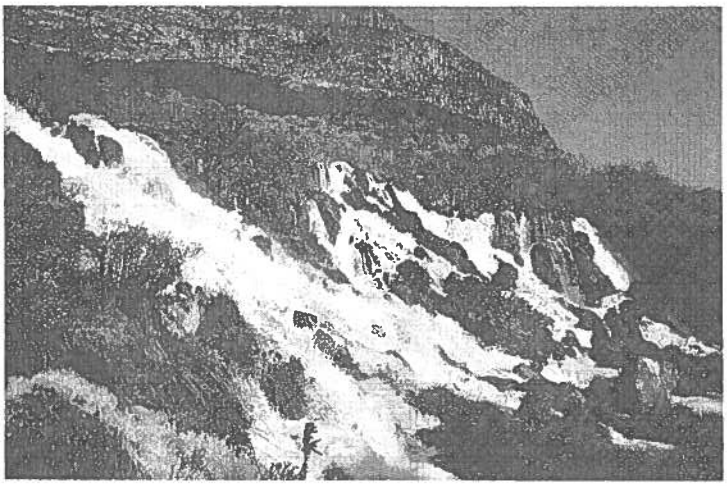


FIGURE 16.23 Thousand Springs, Idaho, emerges from the north side of the Snake River Canyon, opposite the mouth of the Salmon River. The spring issues from a highly permeable layer of scoriaceous basalt. A cliff of columnar basalt can be seen high above the falls. (Jack Williams.)

within the aquifer in response to changes in the amount of recharge received.

Diagram B of Figure 16.22 shows how aquicludes can influence the way in which ground water emerges at the surface. Using the same geological structure as in Diagram A, we have shown a deep river valley cut through the strata. The shale lens is now exposed in the upper wall of the valley. Ground water emerges from the perched water table in a line of springs. A *spring* is any flow of ground water emerging naturally from the solid surface of the earth. (Springs may also emerge from the floors of streams, lakes, or the ocean.) A second line of springs is formed in the lower wall of the valley, where the main water table intersects the surface. Springs of emerging ground water occur under a wide variety of geological conditions and we have shown only one simple case.

Springs are usually insignificant features, going unnoticed because of a concealing cover of vegetation. Many small springs cease to flow in the summer season. Some large springs yield copious, sustained flows of water. An impressive example is the Thousand Springs of the Snake River Canyon in Idaho, where a large volume of flow emerges from highly permeable layers of scoriaceous basalt (Figure 16.23).

Where strata are gently inclined, conditions may be favorable for the development of *artesian flow*, the spontaneous upward movement of ground water in a well or spring. Figure 16.24 shows a dipping *confined aquifer* of sandstone, capped by an aquiclude of shale. The aquifer receives water along its exposed outcrop at a relatively high position and moves to lower levels in the aquifer. Because of hydrostatic pressure, water can rise in wells to a level higher than the upper surface of the aquifer. The limit of rise is the *hydrostatic pressure surface*, shown by a sloping line in Figure 16.24. The elevation of this surface declines with increasing distance from the intake area because energy is expended in the flow of ground water through the aquifer. As the diagram shows, a flowing *artesian well* occurs where the ground surface and wellhead lie below the pressure surface. Where the wellhead lies above the pressure surface, the water will rise only to the level of the pressure surface; it will not emerge as a flowing well. Natural artesian springs can occur where faulting or solution removal has produced a passageway for water to make its way upward through the confining aquiclude.

The word "artesian" is taken from a French word *artésien*, meaning "pertaining to the province of Artois." In about 1750 in this northernmost province of France, deep wells were first drilled to reach confined aquifers and obtain artesian flow. Since then, many regions of sedimentary strata have furnished large amounts of artesian water. One of the most extensive areas producing artesian ground water is the Great Plains region, covering parts of Kansas, Nebraska, and eastern Colorado and extending northward into the region of the Black Hills of South Dakota. Here the Dakota sandstone under-

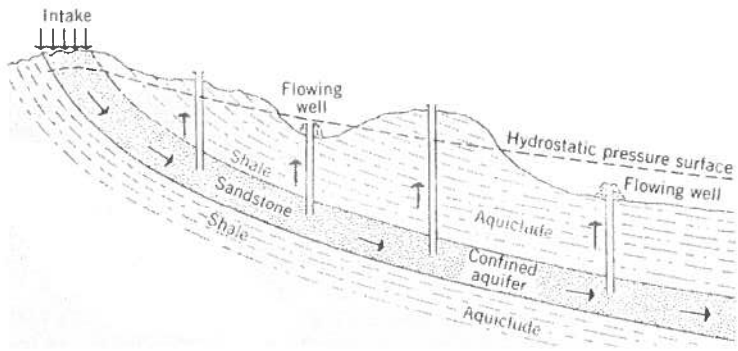


FIGURE 16.24 Schematic cross section of an artesian flow system. (From A. N. Strahler, *Physical Geology*, Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

lies the entire area and has furnished artesian water in many places (see Figure 16.27). Numerous irrigation wells were drilled into the Dakota formation early in this century. Artesian water was withdrawn at a rate that greatly exceeded the rate of recharge. Consequently, few of these wells produce surface flow today and most must be pumped to furnish water.

GROUND-WATER WITHDRAWAL

Withdrawal of ground water for a wide range of human uses has begun to make serious impacts on the environment in many places. The drilling of vast numbers of wells, from which water is forced out in great volumes by powerful pumps, has profoundly altered nature's balance of ground-water recharge and discharge. Increased urban populations and industrial developments require larger water supplies, needs that cannot always be met from construction of new surface-water reservoirs.

In agricultural lands of the semiarid and desert climates, heavy dependence is placed on irrigation water from pumped wells, especially because many of the major river systems have already been fully developed for irrigation from surface supplies. Wells can be drilled within the limits of a given agricultural or industrial property and can provide immediate supplies of water without any need to construct expensive canals or aqueducts.

Formerly, the small well needed to supply domestic and livestock needs of a home or farmstead was actually dug by hand as a large cylindrical hole, lined with masonry where required. By contrast, the modern well put down to supply irrigation and industrial water is drilled by powerful machinery that may bore a hole 40 cm (16 in.) or more in diameter to depths of 300 m (1000 ft) or more. Drilled wells, often called *tube wells*, are sealed off by metal casings that exclude impure near-surface water and prevent clogging of the tube by caving the walls. Near the lower end of the hole, where it enters the aquifer, the casing is perforated to admit the water. The yields of single wells range from as low as a few liters per day in a domestic well to many millions of liters per day for large, deep industrial or irrigation wells.

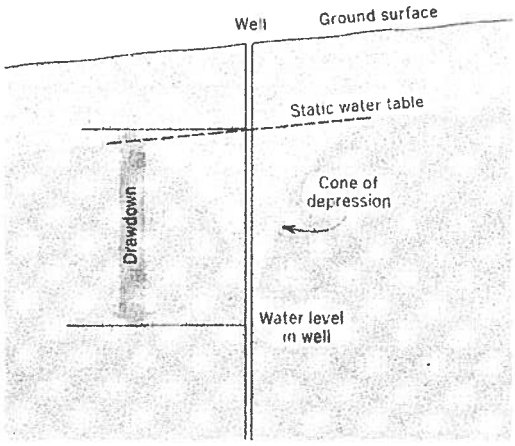


FIGURE 16.25 Drawdown and cone of depression in a pumped well.

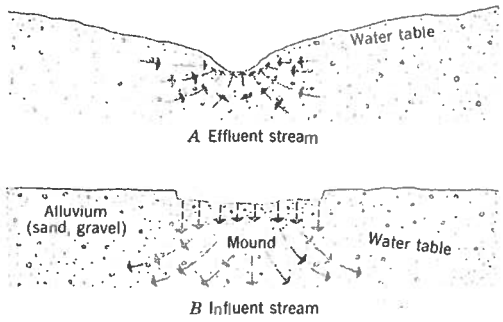


FIGURE 16.26 Effluent and influent streams. (From A. N. Strahler, *The Earth Sciences*, 2nd ed., Harper & Row, Publishers. Copyright © by Arthur N. Strahler.)

As water is pumped from a well, the level of water in the well drops. At the same time, the surrounding water table is lowered in the shape of a conical surface, termed the *cone of depression*. The difference in height between the cone base and the original water table is the *drawdown* (Figure 16.25). By producing a steeper gradient of the water table, the flow of ground water toward the well is also increased, so the well will yield more water. This increase holds only for a limited amount of drawdown, beyond which the yield fails to increase. The cone of depression may extend as far out as 16 km (10 mi) or more from a well where heavy pumping is continued. Where many wells are in operation, their intersecting cones produce a general lowering of the water table.

In areas of moist climates with a large annual water surplus, natural recharge is by general percolation over the entire ground area surrounding the well. Here the prospects of achieving a balance of recharge and withdrawal are highly favorable through the control of pumping. An important recycling measure is the return of waste waters or stream waters to the ground-water table by means of *recharge wells*, in which water flows down rather than up.

Depletion often greatly exceeds the rate at which the ground water of the area is recharged by percolation from rain or from the beds of streams. In an arid region, much of the ground water for irrigation is from wells driven into thick alluvial sands and gravels. Recharge of these deposits depends on the seasonal flows of water from streams originating high in adjacent mountain ranges.

Streams in a dry climate, flowing on plains underlain by sand and gravel, lose water by seepage through the channel floor. This water recharges the ground-water body below and causes the water table to be raised in the form of a mound (Figure 16.26*A*). Streams of this type are called *influent streams*. In contrast, in moist climates where the water table is high and slopes toward the channels, streams receive flow by out-seepage of ground water; these are *effluent streams* (Figure 16.26*B*).

In dry climates, particularly, the extraction of ground